Introduction to statistics

(Day 2)

Recap

Recap

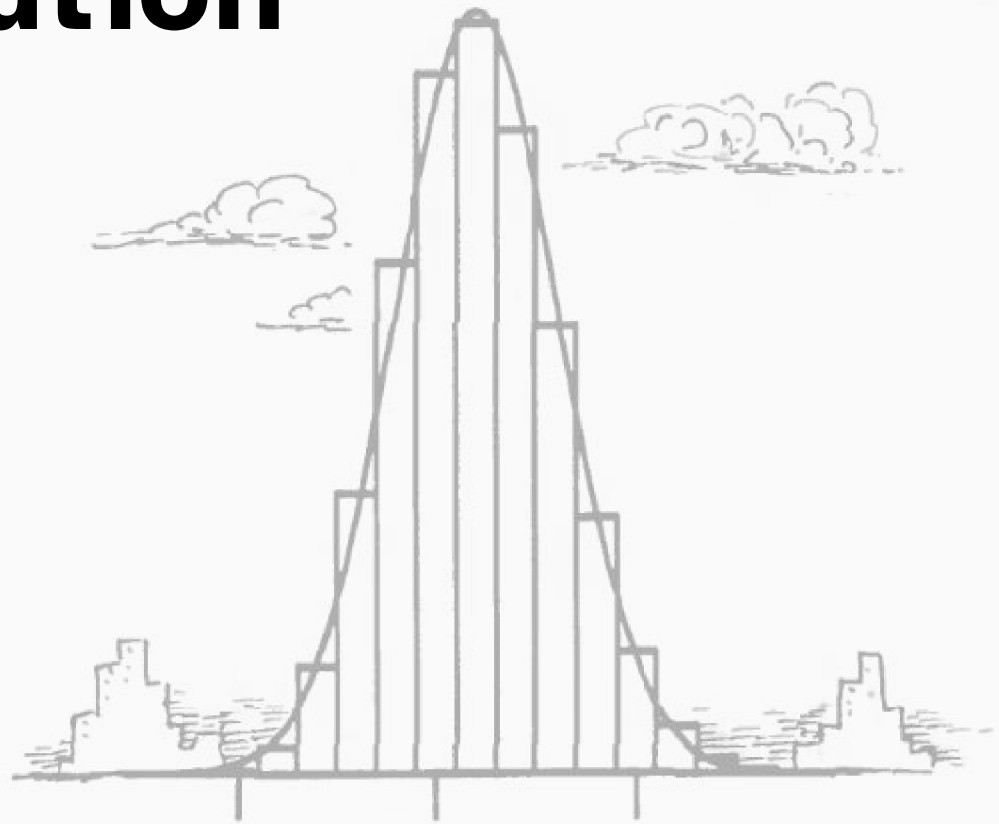
- The collection, organisation, summarisation, and analysis of data
  → *Descriptive* statistics
- The drawing of inferences about a body of data when only a part of the data is observed
  → *Inferential* statistics

- When can't study a population, we select a representative sample
- There are different sampling strategies

Recap

- There are different types of variables
- Categorical variables are described with absolute and relative frequencies
- Numerical variables are described with measures of central tendency, dispersion, and correlation
- Variables can be summarised with multiple graphical representations
- Parameters (calculated on the population) *vs* statistics (calculated on the sample)

The Normal Distribution



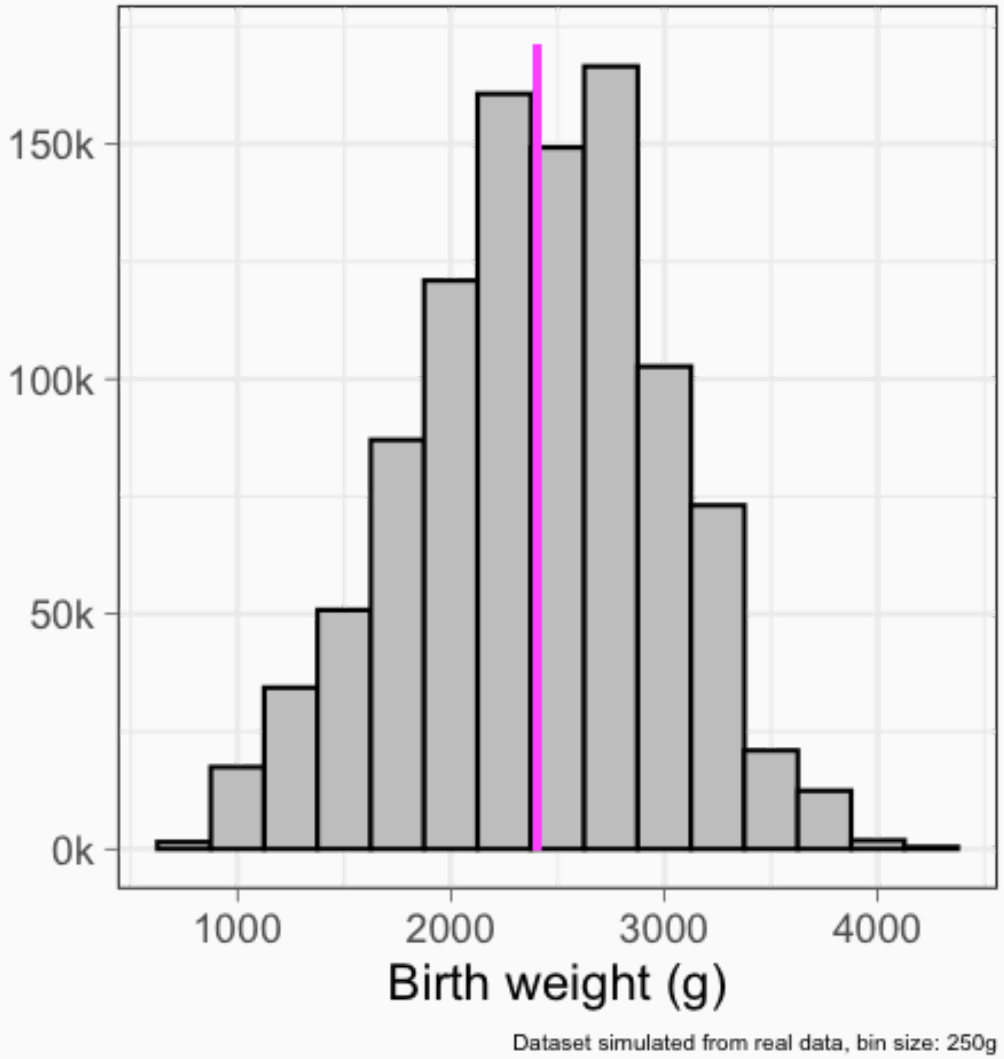
Learning objectives

- Understand the characteristics of the Normal and the Standard Normal distributions
- Calculate and interpret z-scores
- Determine the proportion of individuals in a population who exhibits a certain characteristic
- Calculate the probability that an individual in a population exhibits a certain characteristic

Population distribution

What is the distribution of birth weight among English twins?

$$N = 1\text{M}$$
$$median = 2400 \text{ g}$$
$$\mu = 2400 \text{ g}; \ \sigma = 580 \text{ g}$$



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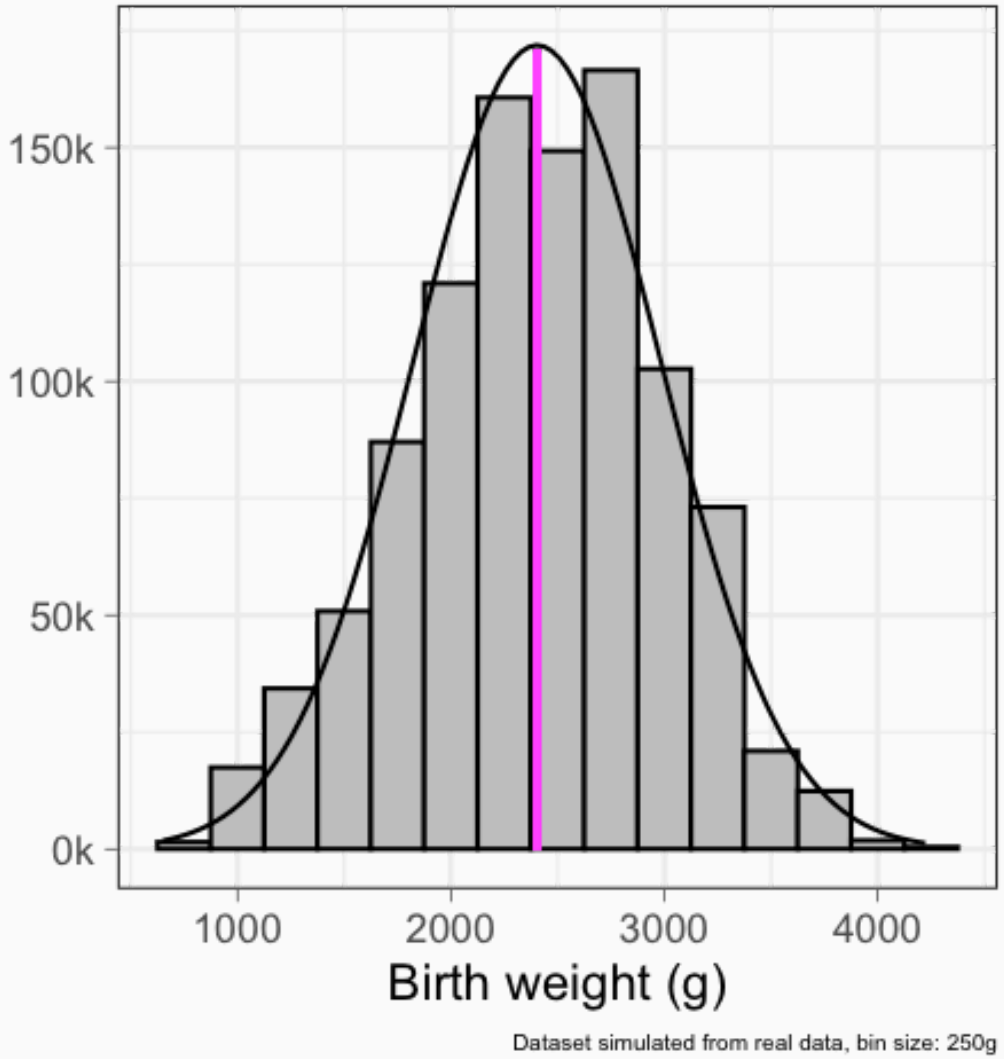
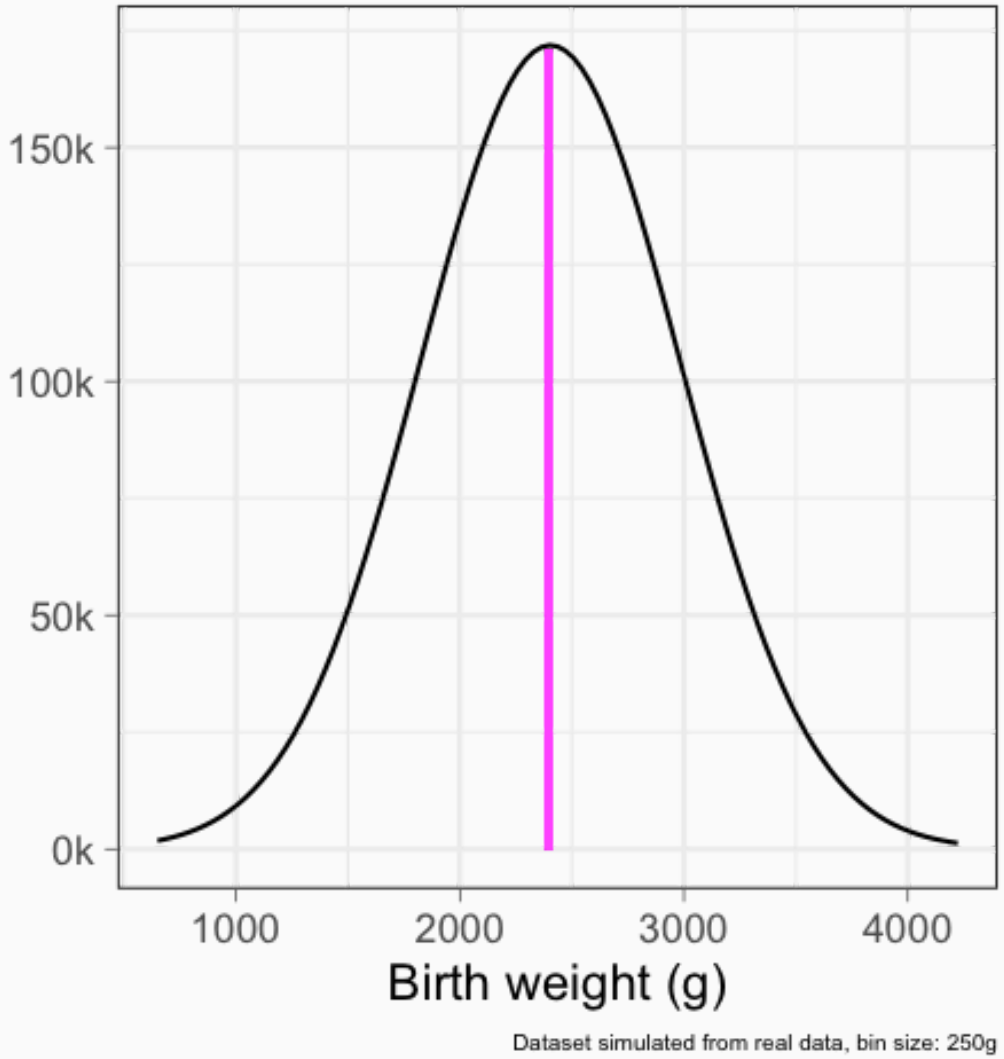


The Normal distribution

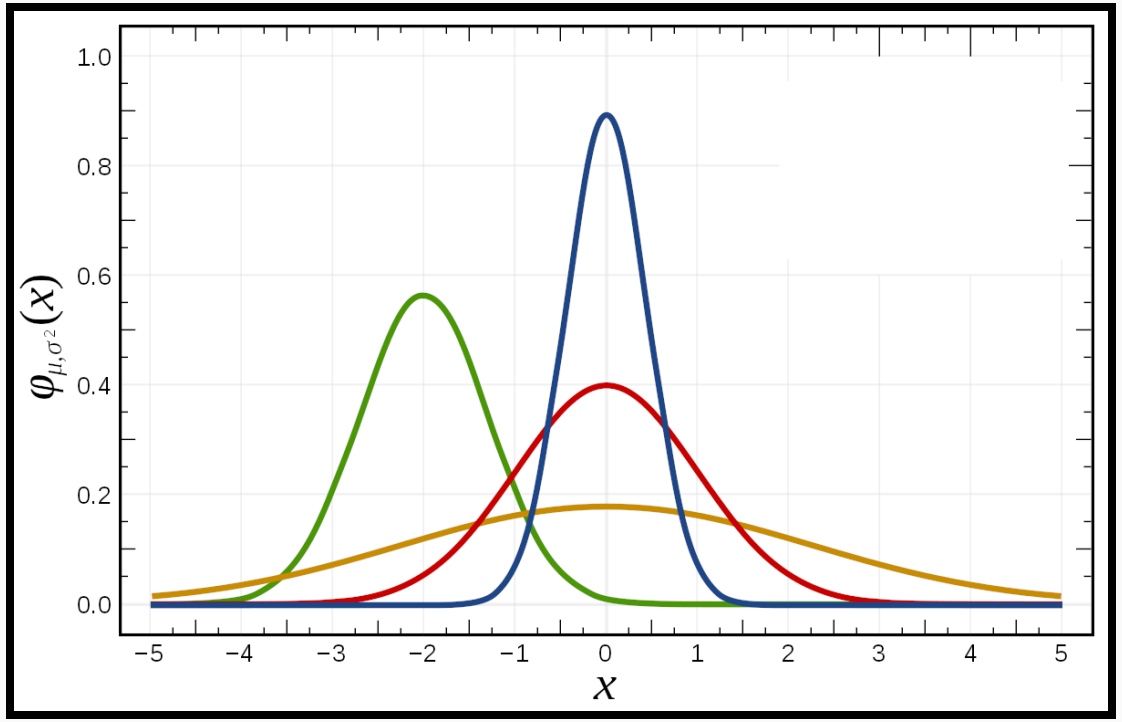
- $\mathcal{N} = (\mu, \sigma^2)$
- $\text{mode} \equiv \text{mean} \equiv \text{median}$
- Symmetrical



Exercise #1

? Which curve has the largest mean?

a) Green
b) Blue
c) Yellow
d) None of the above



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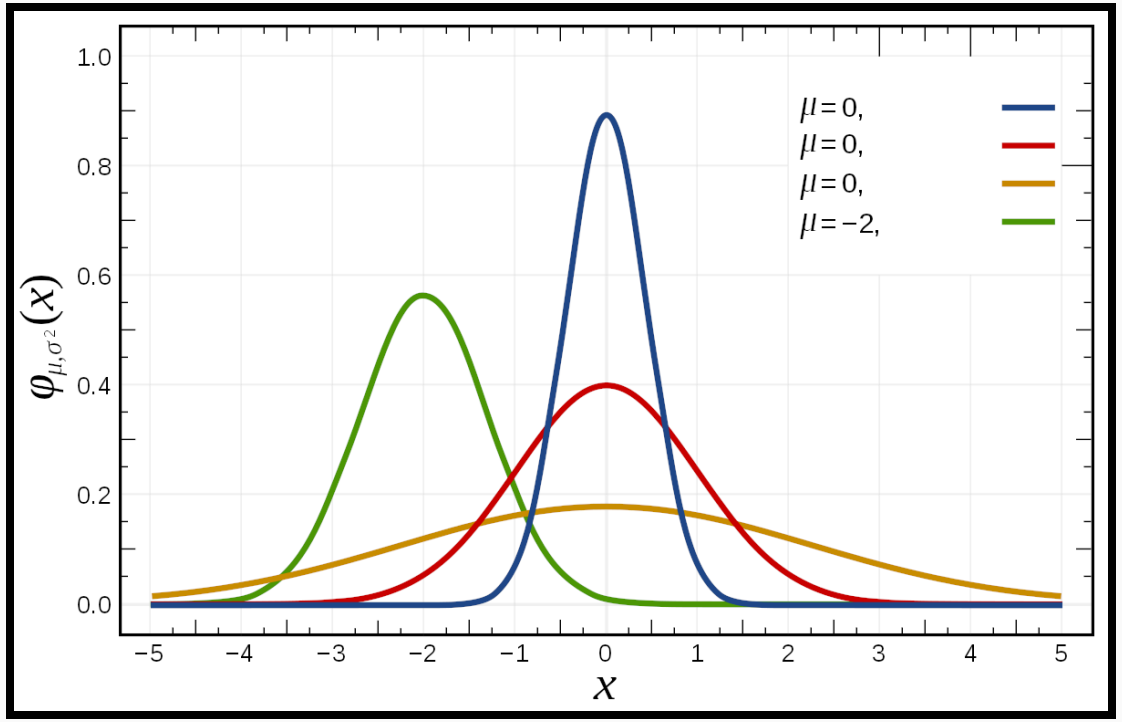
Exercise #1 -- Solution

❓ Which curve has the largest mean?

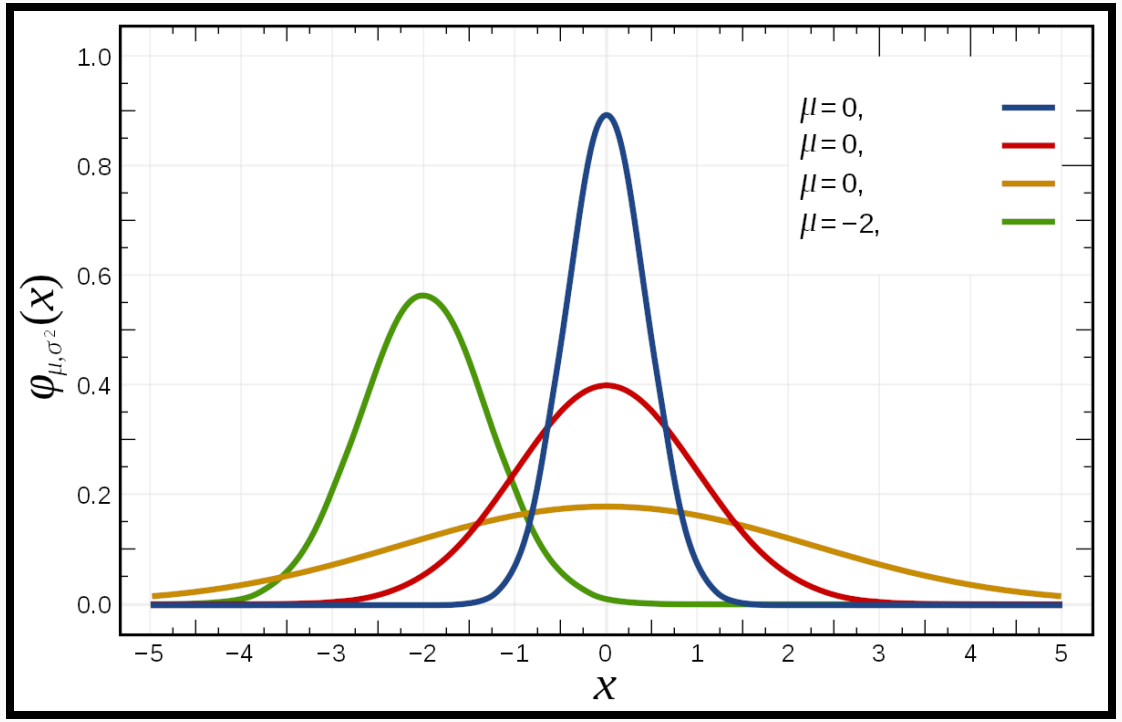
a) Green
b) Blue
c) Yellow
d) None of the above ✅



Exercise #2

? Which curve has the largest standard deviation?

a) Green
b) Blue
c) Yellow
d) None of the above



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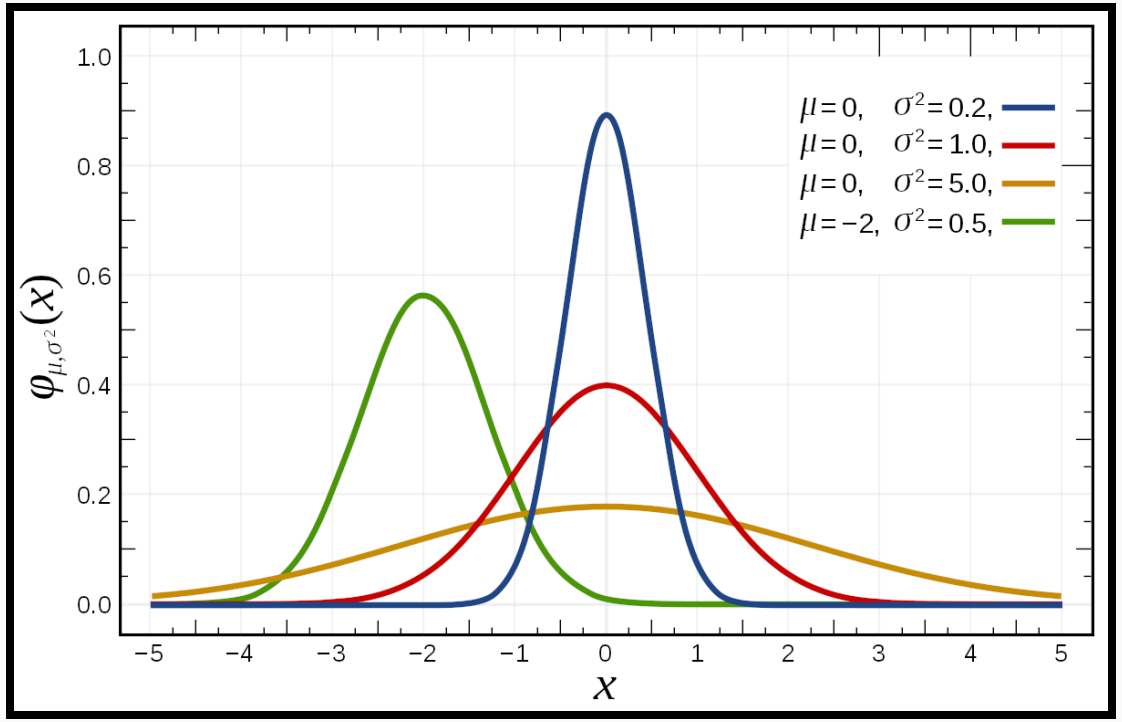
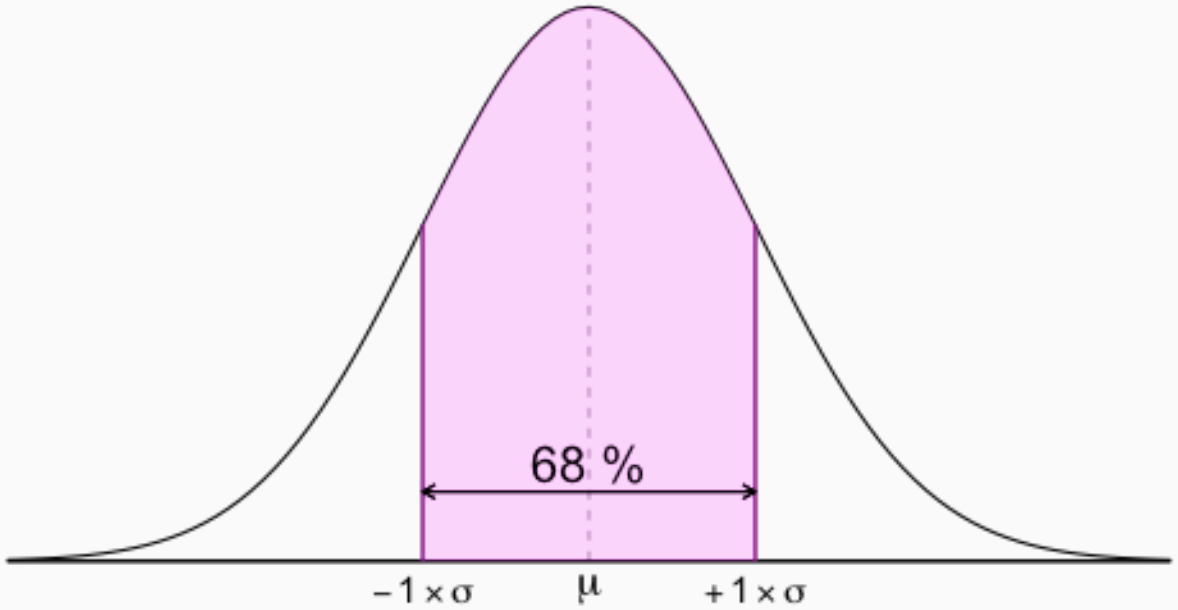


Three-sigma rule

- approximately 68% of the values lie within one standard deviations (1 σ) of the mean



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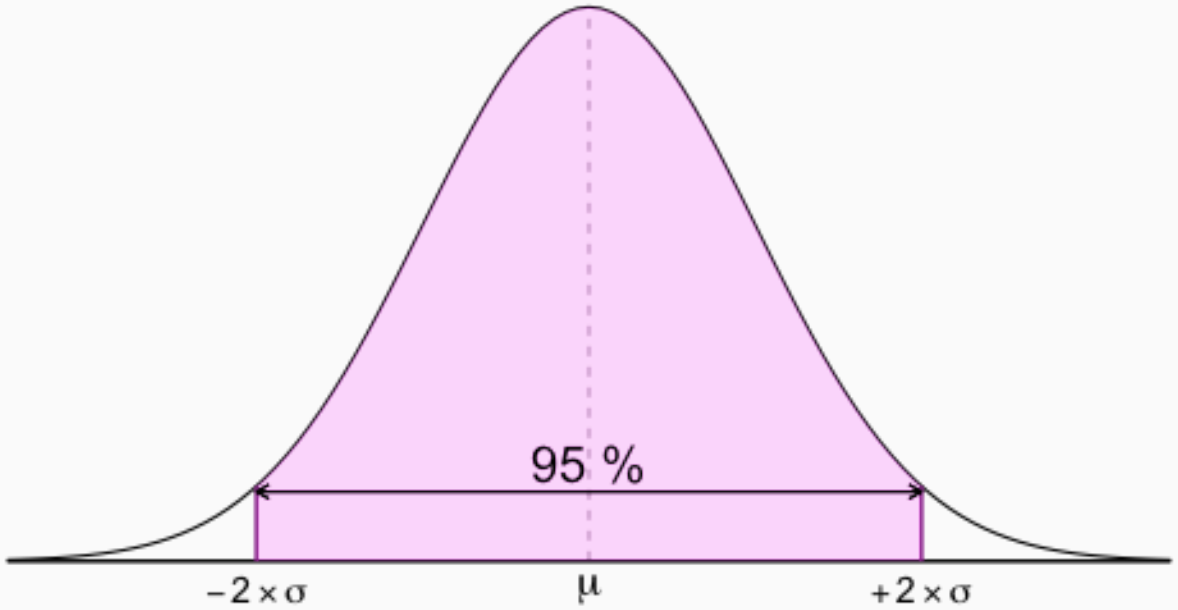

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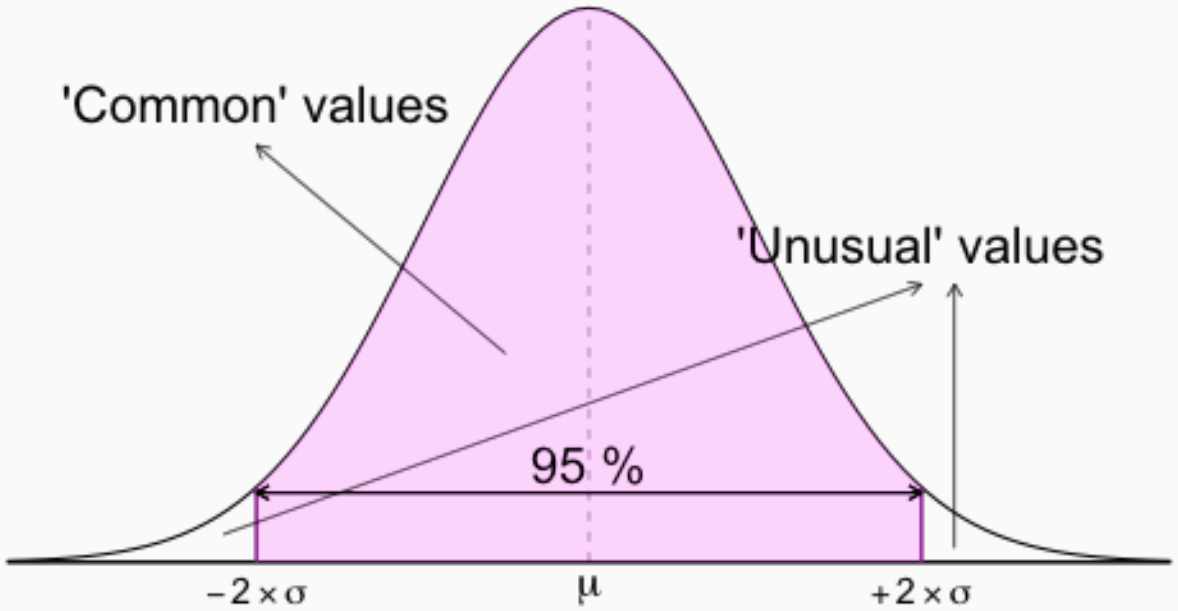

Three-sigma rule

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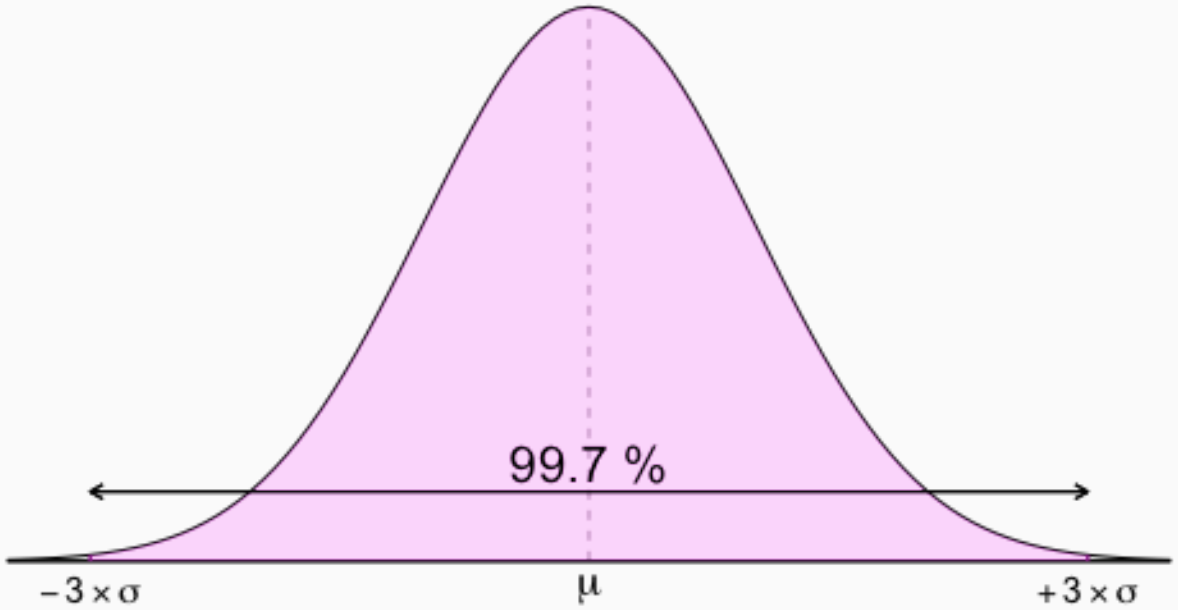
- approximately 95% lie within 2 σ
- approximately 99.7% lie within 3 σ



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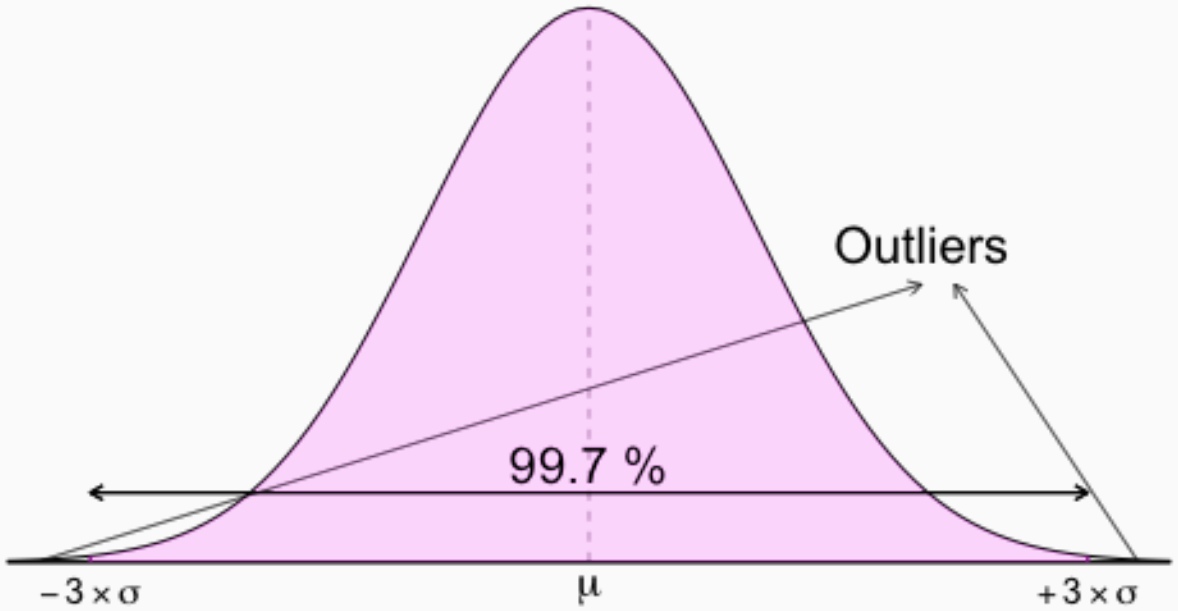
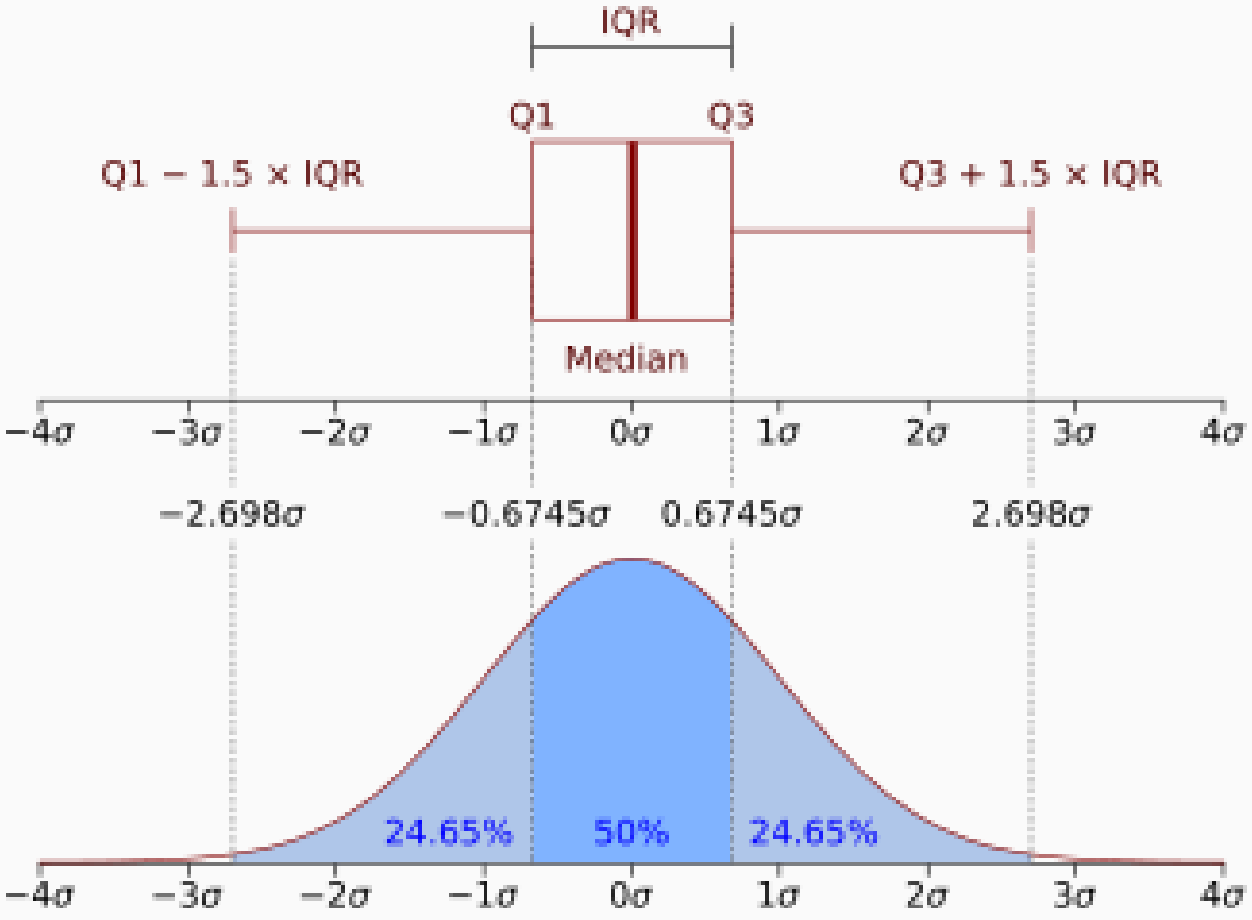


Back to the boxplot.



Exercise #3

? The height of the Italian male population follows a Normal distribution with a mean of 170 cm and a standard deviation of 9.5 cm.

Calculate the following values, when possible

a) The median height
b) The proportion of Italian men taller than $170\ \mathrm{cm}$
c) The values considered unusual and/or outliers
d) The most common height
e) The range that includes 68% of the individuals
f) The height of the tallest Italian man

02:00

Exercise #3 -- Solution

? The height of the Italian male population follows a Normal distribution with a mean of 170 cm and a standard deviation of 9.5 cm.

Calculate the following values, when possible

a) The median height $\rightarrow$ coincide with the mean $= 170\ \mathrm{cm}$
b) The proportion of Italian men taller than $170\ \mathrm{cm}$

Exercise #3 -- Solution

? The height of the Italian male population follows a Normal distribution with a mean of 170 cm and a standard deviation of 9.5 cm.

Calculate the following values, when possible

a) The median height $\rightarrow$ 170cm

b) The proportion of Italian men taller than $170\ \mathrm{cm} \rightarrow$ those at the right of the median, half of the area under the curve $= 50\%$

c) The values considered unusual and/or outliers

Exercise #3 -- Solution

? The height of the Italian male population follows a Normal distribution with a mean of 170 cm and a standard deviation of 9.5 cm.

Calculate the following values, when possible

a) The median height $\rightarrow$ 170cm
b) The proportion of Italian men taller than $170\ \mathrm{cm} \rightarrow 50\%$
c) The values considered unusual and/or outliers $\rightarrow$ those > 2 standard deviation from the mean $= 170 - 2 \times 9.5 = 151\ \mathrm{cm}$ and $170 + 2 \times 9.5 = 189\ \mathrm{cm}$
d) The most common height

Exercise #3 -- Solution

? The height of the Italian male population follows a Normal distribution with a mean of 170 cm and a standard deviation of 9.5 cm.

Calculate the following values, when possible

a) The median height $\rightarrow$ 170cm
b) The proportion of Italian men taller than $170\ \mathrm{cm} \rightarrow 50\%$
c) The values considered unusual and/or outliers
$\rightarrow \ < 151\ \mathrm{cm}\ \mathrm{and}\ > 189\ \mathrm{cm}$
d) The most common height $\rightarrow$ the mode, which coincides with the mean and the median $= 170\ \mathrm{cm}$
e) The range that includes 68% of the individuals

Exercise #3 -- Solution

? The height of the Italian male population follows a Normal distribution with a mean of 170 cm and a standard deviation of 9.5 cm.

Calculate the following values, when possible

a) The median height $\rightarrow$ 170cm
b) The proportion of Italian men taller than $170\ \mathrm{cm} \rightarrow 50\%$
c) The values considered unusual and/or outliers
$\rightarrow \quad < 151\ \mathrm{cm}\ \mathrm{and} > 189\ \mathrm{cm}$
d) The most common height $\rightarrow 170\ \mathrm{cm}$
e) The range that includes 68% of the individuals $\rightarrow$ that included between ± 1 standard deviation from the mean
$= 170 - 9.5\ \mathrm{to}\ 170 + 9.5\ \mathrm{cm}\ = 160.5\ \mathrm{to}\ 179.5$
f) The height of the tallest Italian man

Exercise #3 -- Solution

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Calculate the following values, when possible

a) The median height $\rightarrow$ 170cm
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$\rightarrow \quad < 151 \text{ cm and} > 189 \text{ cm}$
d) The most common height $\rightarrow 170 \text{ cm}$
e) The range that includes 68% of the individuals $\rightarrow 160.5 \text{ to } 179.5 \text{ cm}$
f) The height of the tallest Italian man $\rightarrow$ we can't calculate this!

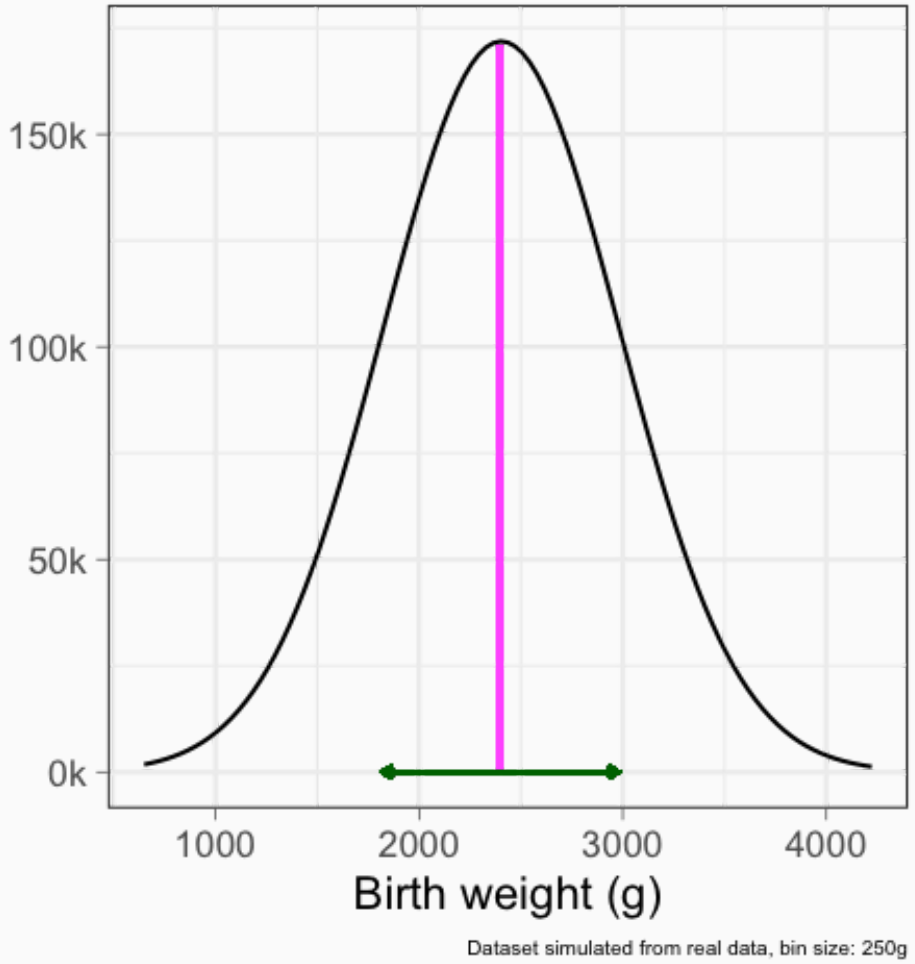
Characterize a single observation

Suppose we are caring for a newborn twin who weighs 1450g.

How can this newborn's weight be characterized in relation to the overall population?

Let’s take a step back…

- The mean indicates the center of a distribution
- The standard deviation indicates the "typical" distance from the mean



Characterize a single observation

Suppose we are caring for a newborn twin who weighs 1450g

- The mean indicates the center of a distribution

$$x = 1450 \text{ g} < \mu = 2400 \text{ g} \quad \rightarrow \quad x - \mu = 1450 \text{ g} - 2400 \text{ g} = -950 \text{ g}$$

$\rightarrow$ The newborn weighs below the mean

Characterize a single observation

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- The standard deviation indicates the "typical" distance from the mean

$$|x - \mu| = 950 \text{ g} > \sigma = 580 \text{ g}$$

$\rightarrow$ The weight is "atypically" far from the mean

Characterize a single observation

Suppose we are caring for a newborn twin who weighs 1450g

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$$x = 1450\ \mathrm{g} < \mu = 2400\ \mathrm{g} \quad \rightarrow \quad x - \mu = 1450\ \mathrm{g} - 2400\ \mathrm{g} = -950\ \mathrm{g}$$

$\rightarrow$ The newborn weighs below the mean

- The standard deviation indicates the "typical" distance from the mean

$$|x - \mu| = 950\ \mathrm{g} > \sigma = 580\ \mathrm{g} \quad \rightarrow \quad \frac{x-\mu}{\sigma} = \frac{-950\ \mathrm{g}}{580\ \mathrm{g}} = -1.87$$

$\rightarrow$ The weight is "atypically" far from the mean

$\rightarrow$ It is an (almost) "unusual" weight

The z-score

$$z = \frac{x - \mu}{\sigma}$$

- Indicates whether an observation is above or below the population mean
- Indicates whether an observation's deviation from the mean is large or small relative to the population's typical deviation

Exercise #4

? Maria suffered a traumatic brain injury following an accident, and the neurologist caring for her administered three tests.

1. Memory test: Maria heard words and had to repeat them. She recalled 6 words; the general population recalls a mean of 7 words, with a standard deviation of 1.3 words.

2. Object naming test: Maria had to identify objects from drawings. She recognized 7; the general population recognizes a mean of 10 objects, with a standard deviation of 0.59 objects.

3. Stroop test: Maria was given a list of colours written in different inks and had to name the ink colour as quickly as possible. She took 15.7 seconds; the general population takes a mean of 16.2 seconds, with a standard deviation of 1.3 seconds.

Based on these results, should the neurologist focus on Maria's memory, object naming ability, or attention in future visits?

03:00

Exercise #4 -- Solution

? Maria suffered a traumatic brain injury following an accident, and the neurologist caring for her administered three tests.

1. Memory test. $x = 6$; $\mu = 7$, $\sigma = 1.3$
   $z = \frac{6-7}{1.3} = -0.77$

2. Object naming test. $x = 7$; $\mu = 10$, $\sigma = 0.59$
   $z = \frac{7-10}{0.59} = -5.09$

3. Stroop test. $x = 15.7$; $\mu = 16.2$, $\sigma = 1.3$
   $z = \frac{15.7-16.2}{1.3} = -0.39$

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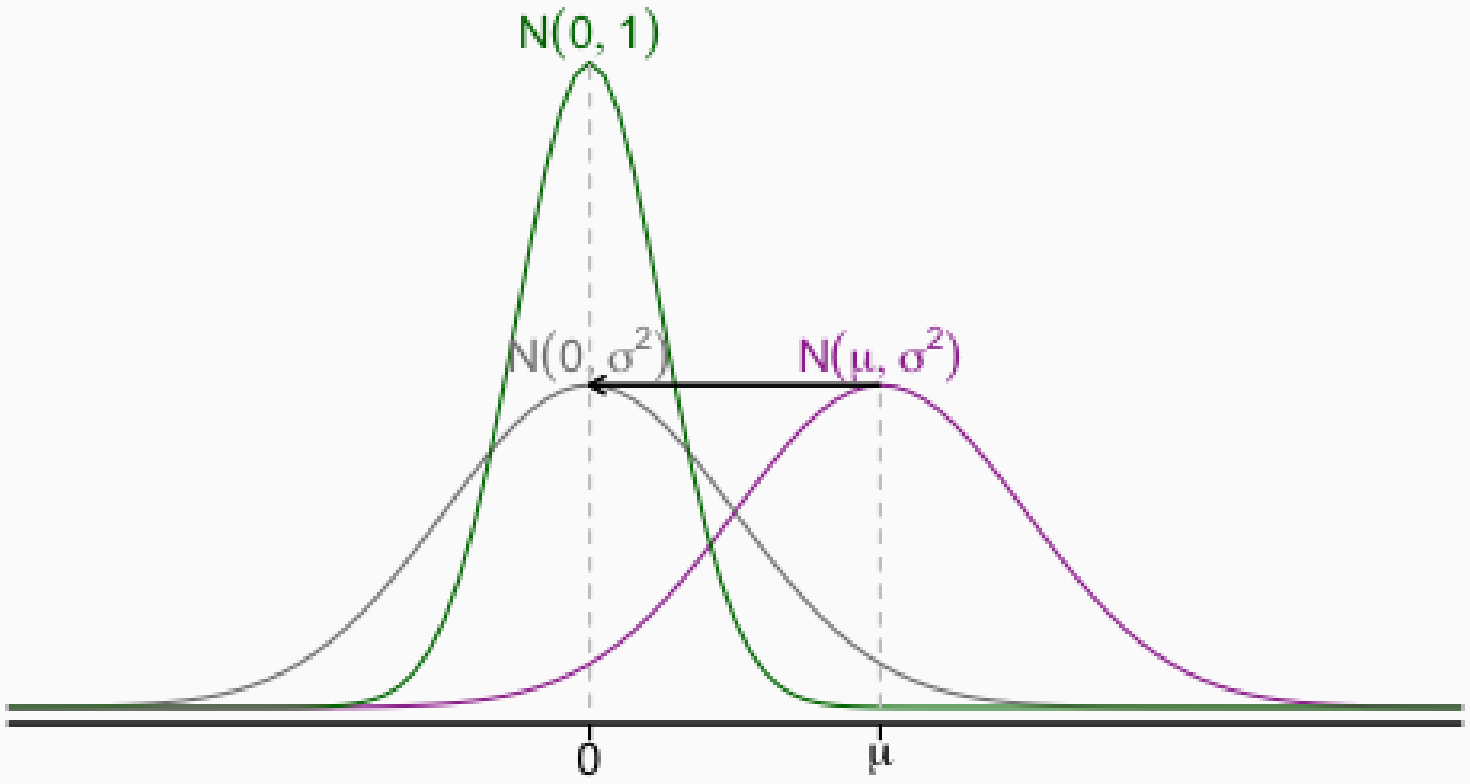
Based on these results, should the neurologist focus on Maria's memory, object naming ability, or attention in future visits?

$\rightarrow$ object naming ability

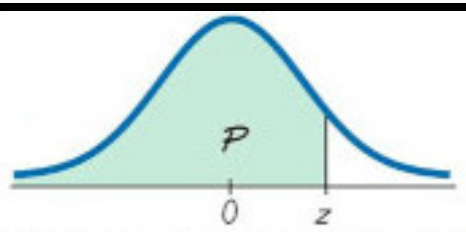
The Standard Normal distribution

- $z = \frac{x-\mu}{\sigma}$
- $\mathcal{N} = (\mu, \sigma^2) \rightarrow Z = (0, 1)$



The Standard Normal distribution

- $Z = (0, 1)$
- Area under the curve. $= 1$
- proportion $\equiv$ probability

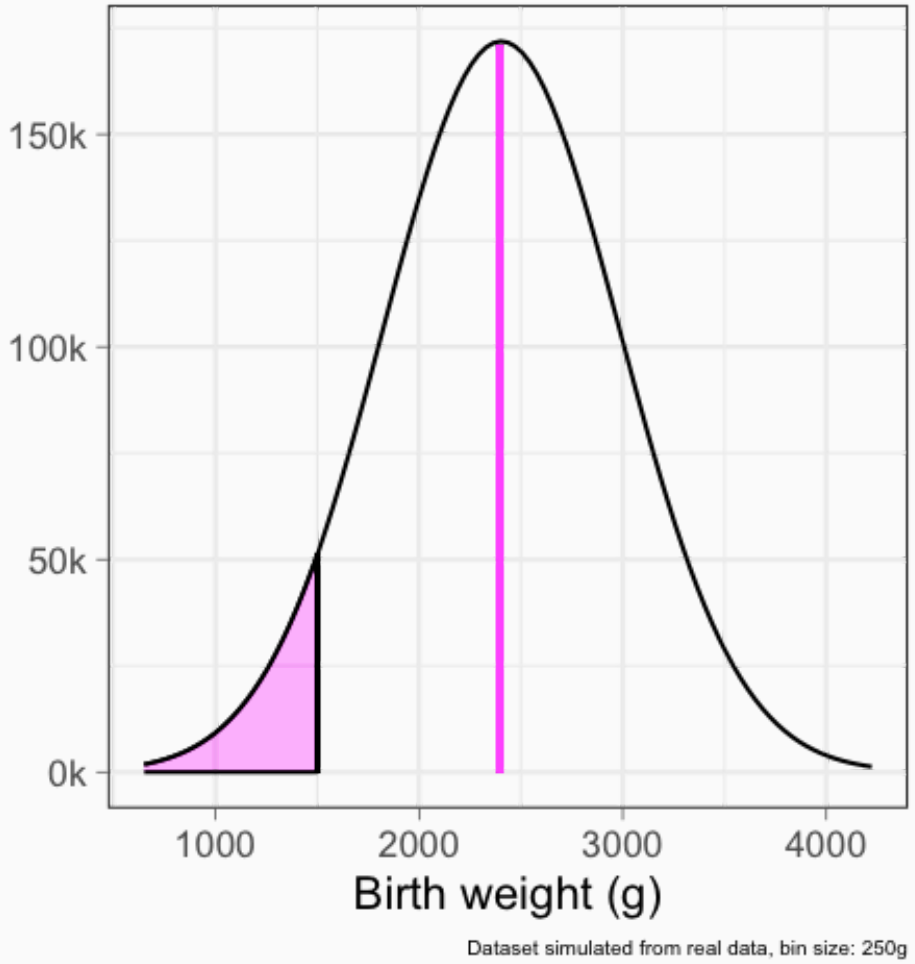


z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441

Proportion $\equiv$ probability

- 6% of twins are born with very low birth weight (< 1500g)
- The probability of being born with very low birth weight is 0.06

But how was it calculated?



Let’s calculate the probability/proportion.

What is the probability that a twin is born with very low birth weight?

$$\mathcal{N} = (2400, 580^2)$$

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1. Compute the z-score

$$z = \frac{x-\mu}{\sigma} = \frac{1500-2400}{580} = \frac{-900}{580} = -1.55$$

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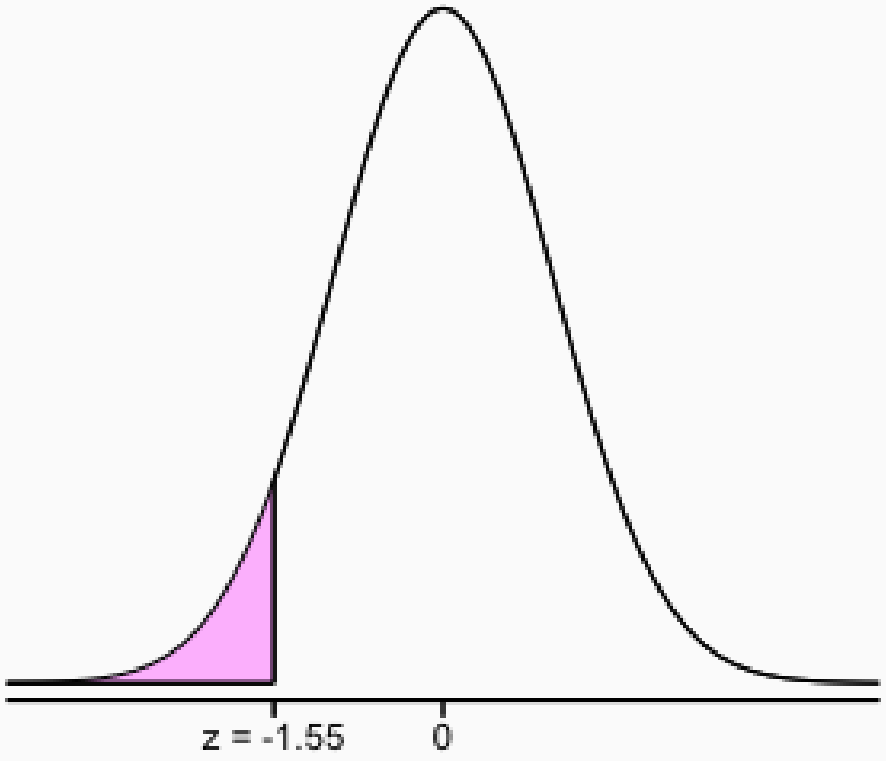
2. Draw a Normal Distribution, and
   shade the area we are interested in



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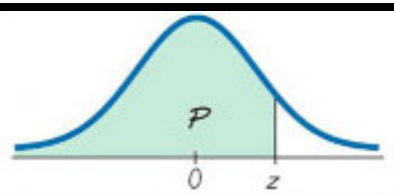
2. Draw a Normal Distribution, and shade the area we are interested in

3. Use a Unit Normal Table to find the area of the shaded curve



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1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441

Let's calculate the probability/proportion.

What is the probability that a twin is born with very low birth weight?

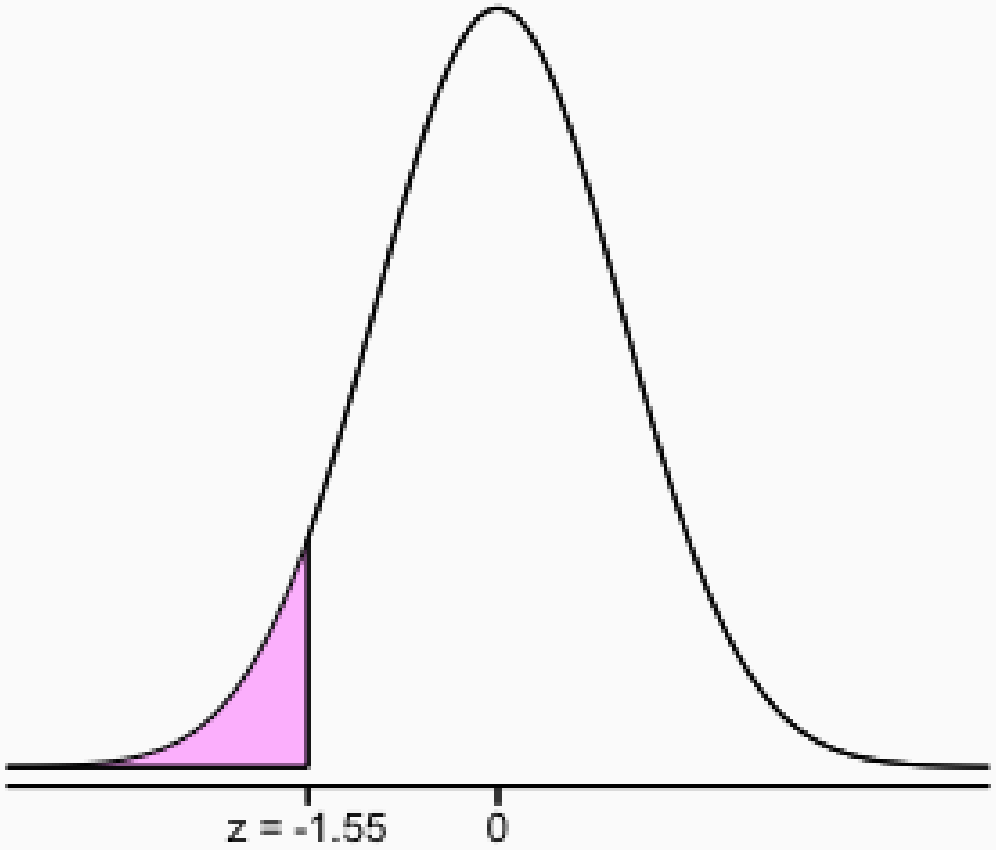
$$\mathcal{N} = (2400, 580^2)$$

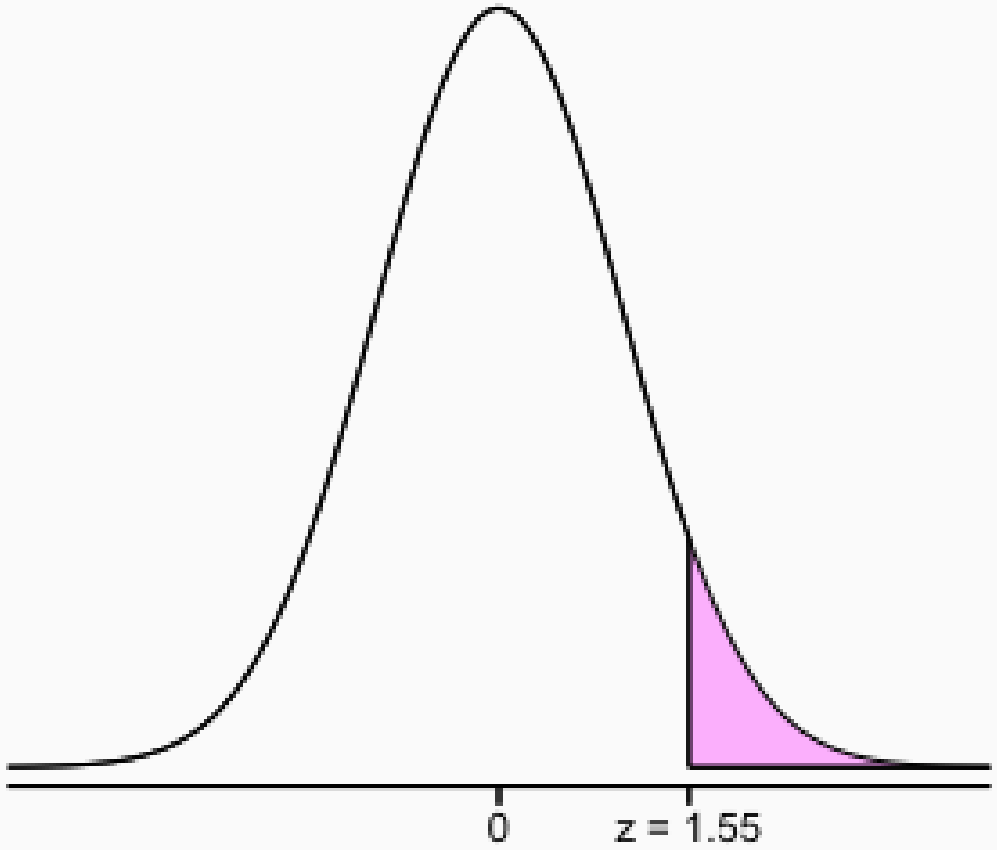
1. Compute the z-score
   $z = \frac{x-\mu}{\sigma} = \frac{1500-2400}{580} = \frac{-900}{580} = -1.55$

2. Draw a Normal Distribution, and
   shade the area we are interested in

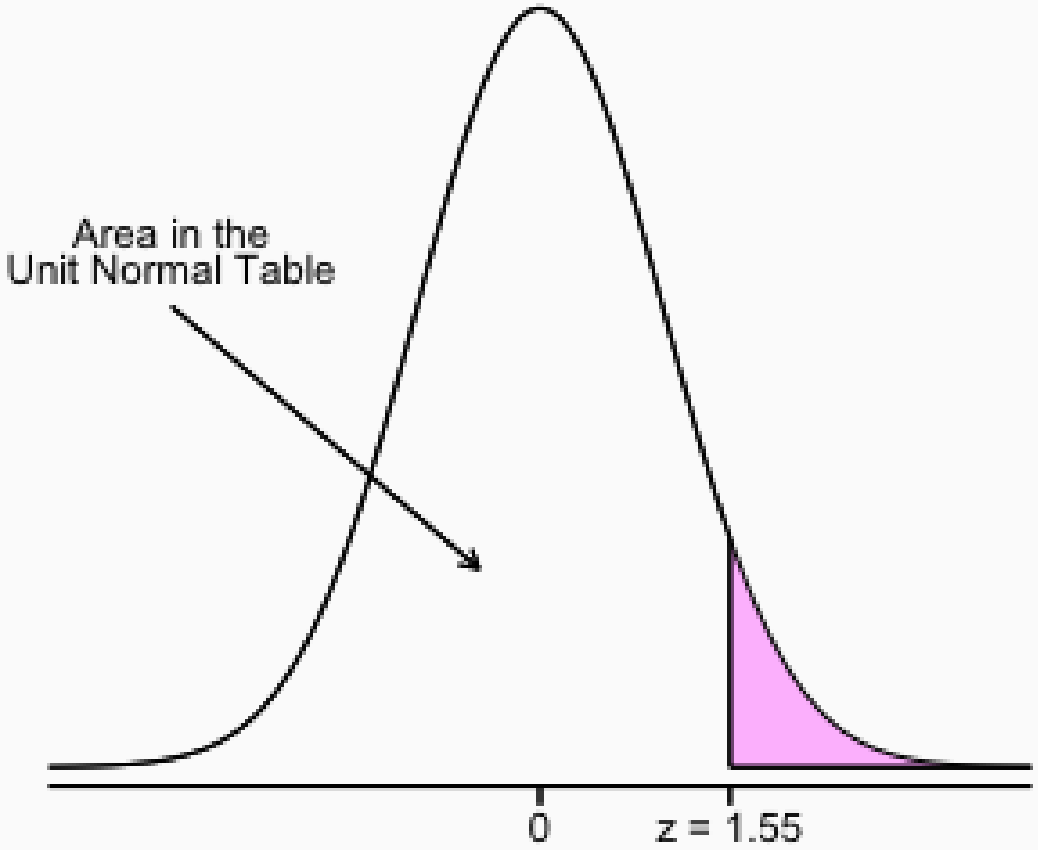
3. Use a Unit Normal Table to find
   the area of the shaded curve
   $\rightarrow$ There are no negative z-scores

Find the area of the shaded curve





Find the area of the shaded curve



Compute the probability/proportion

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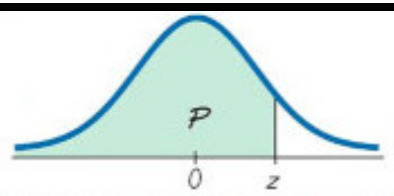
$$\mathcal{P} = 1 - 0.9394 = 0.0606 \rightarrow 6.06\%$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
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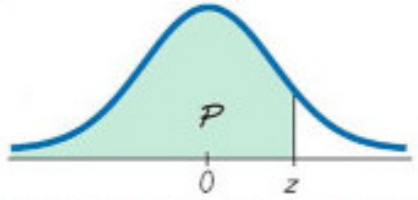
Exercise #5

? Not knowing that the baby is a twin, the pediatrician tells the mother that a birth weight below 2500 g is unusual. Is it really unusual?

$$\mu = 2400 \text{ g}; \ \sigma = 580 \text{ g}$$

05:00



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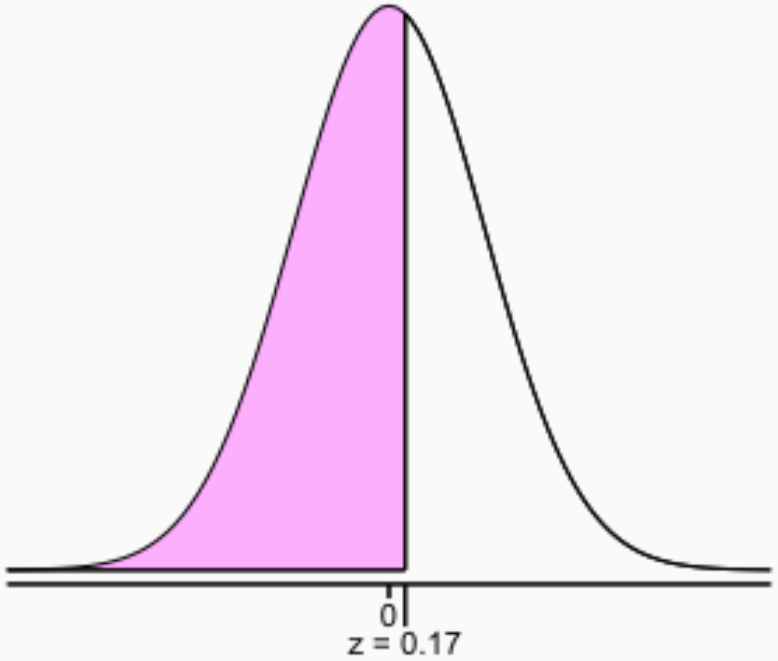

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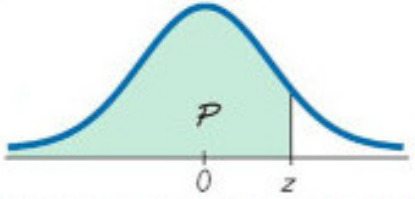
$$\mathcal{P}(x < 2500) = 0.5675 \rightarrow 56.75\%$$



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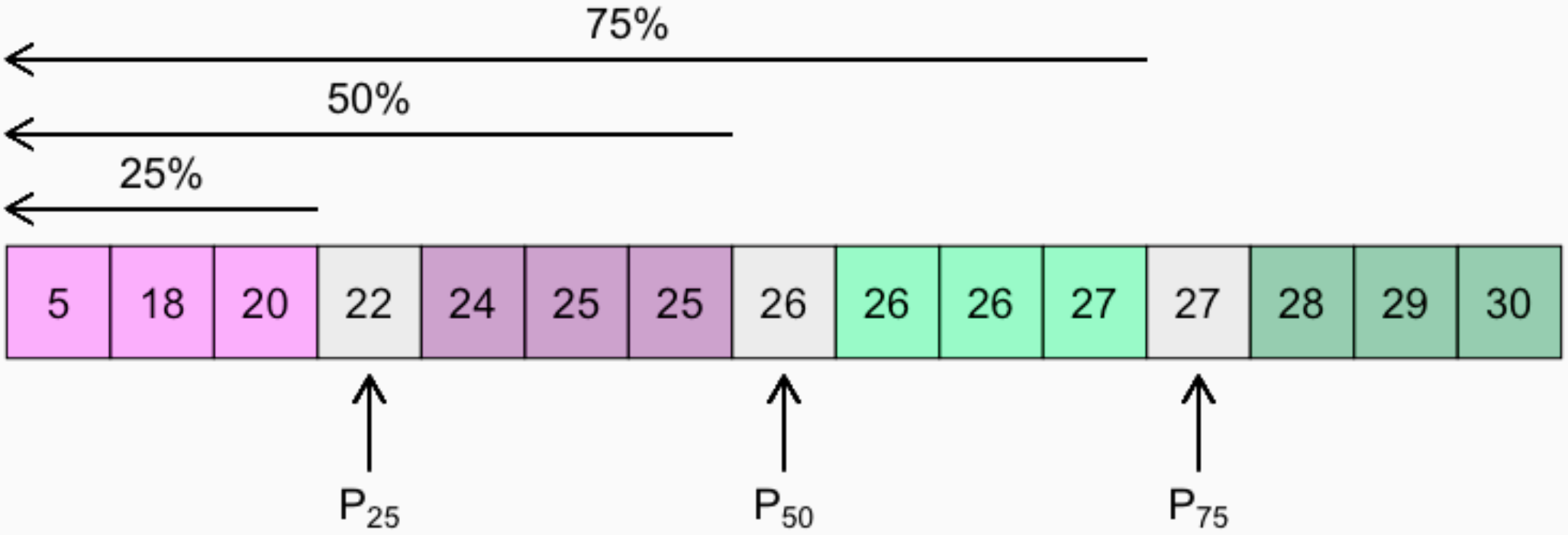
Percentiles

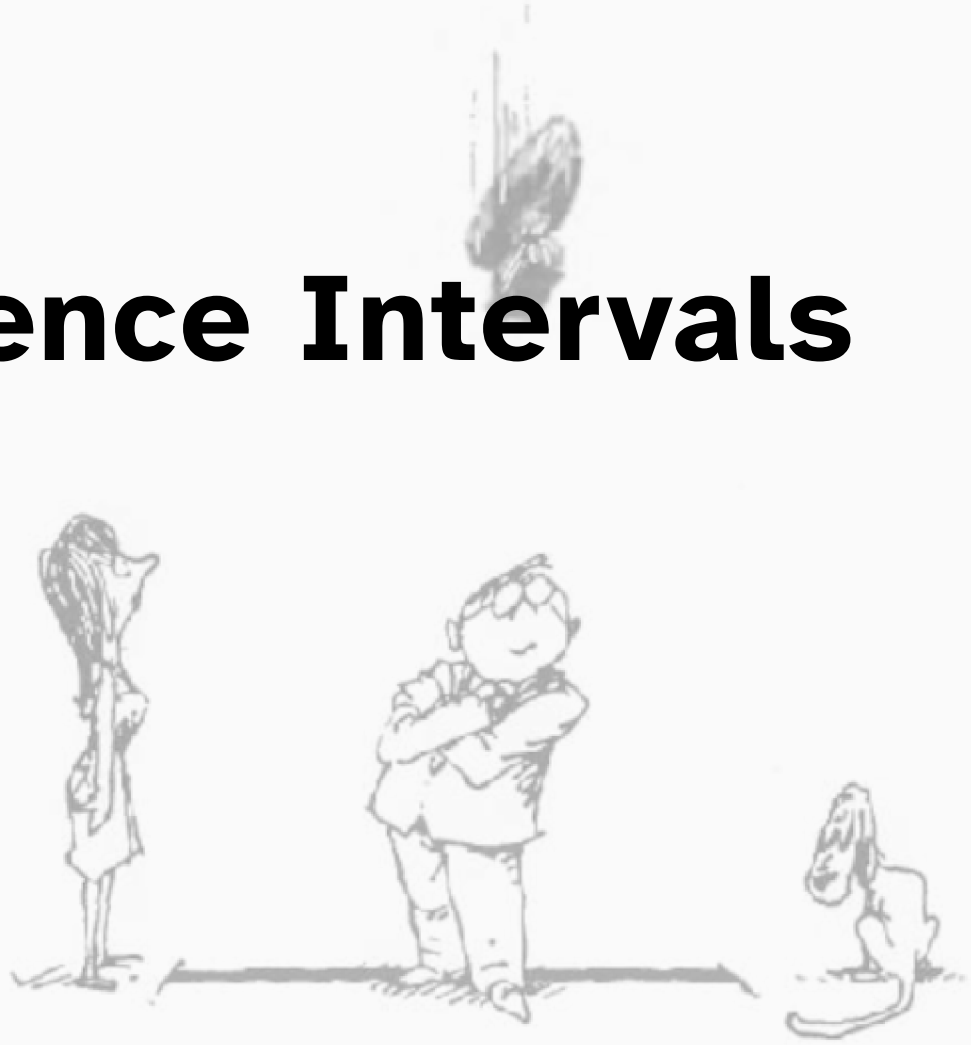


$\mathcal{P} = 0.57$ tells us that our newborn twin is in the 57th percentile

Summary

- Many natural phenomena follow a Normal distribution
- The Normal distribution is defined by the mean and standard deviation and corresponds to a probability distribution
- A population's Normal distribution allows us to determine both the probability of observing a certain value and its expected frequency
- If the data follow a Normal distribution, approximately 68% of values lie within 1 standard deviation of the mean, 95% within 2, and 99.7% within 3
- The z-score allows us to position an observation relative to the reference population and to compare data from very different distributions

Estimates and Confidence Intervals



Learning objectives

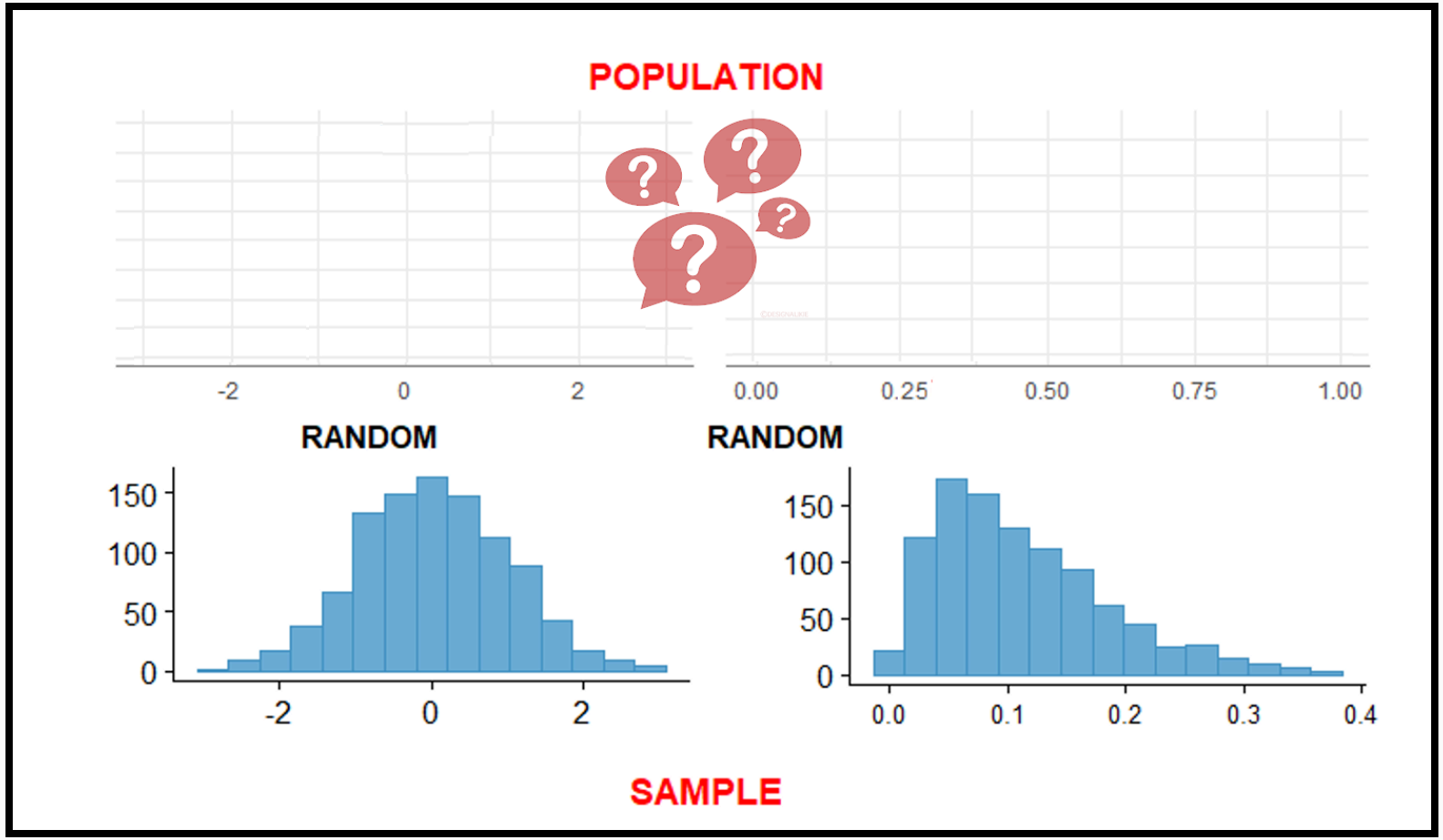
- Be able to use sample statistics to infer population parameters
- Be able to communicate the uncertainty of a statistic
- Be able to calculate and interpret a confidence interval

⚠️ Disclaimer ⚠️

If this part seems complex, it’s because it really is.

You may need to spend some time fully understanding it
Don’t worry, we’ve all been there!

From sample to population



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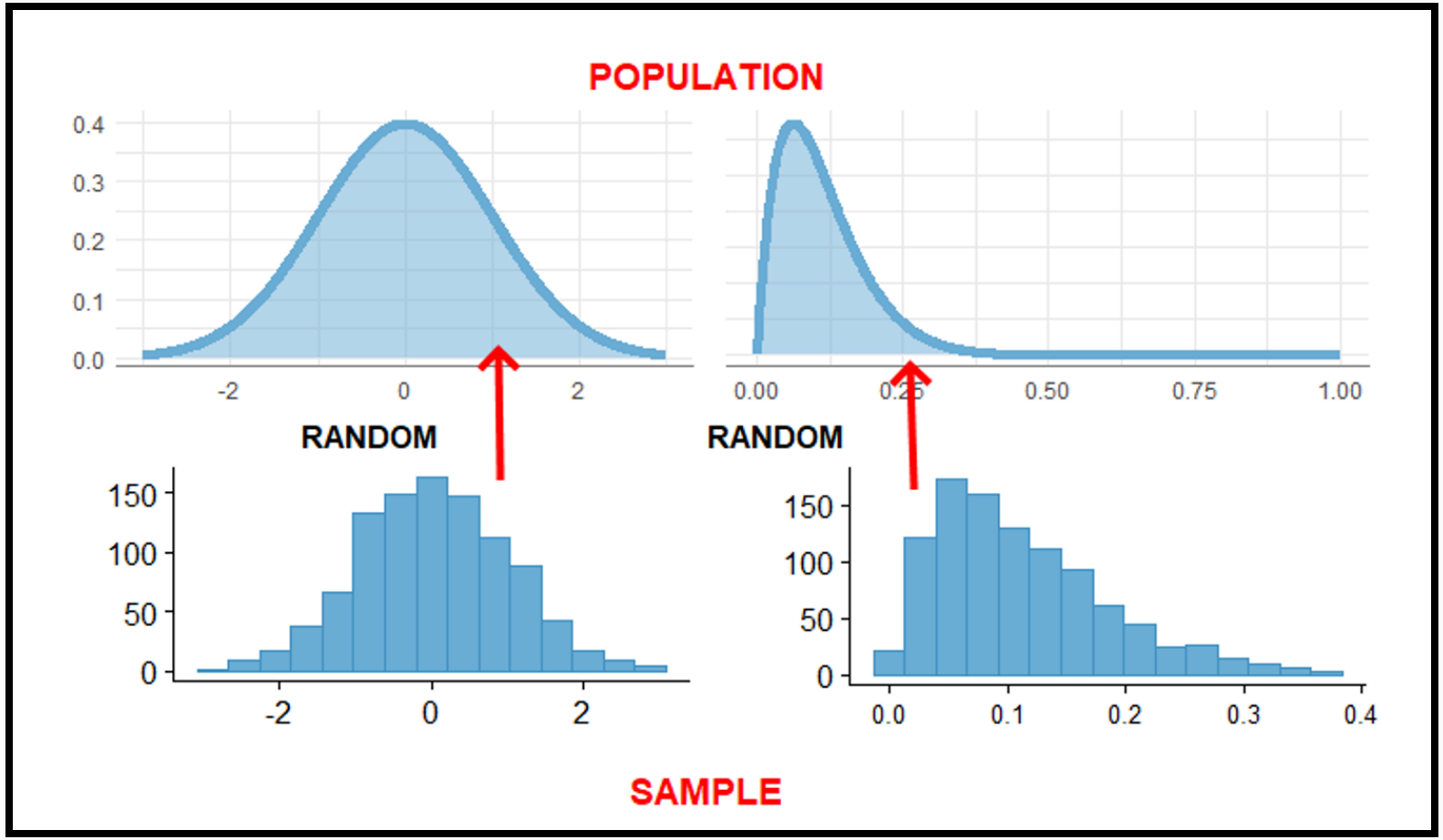
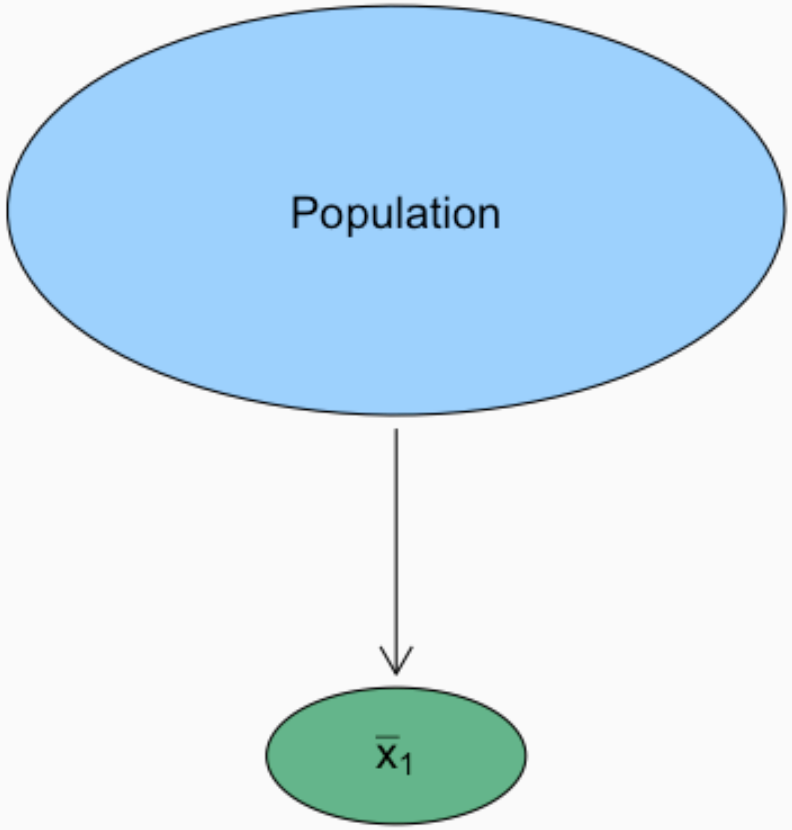


Sampling error



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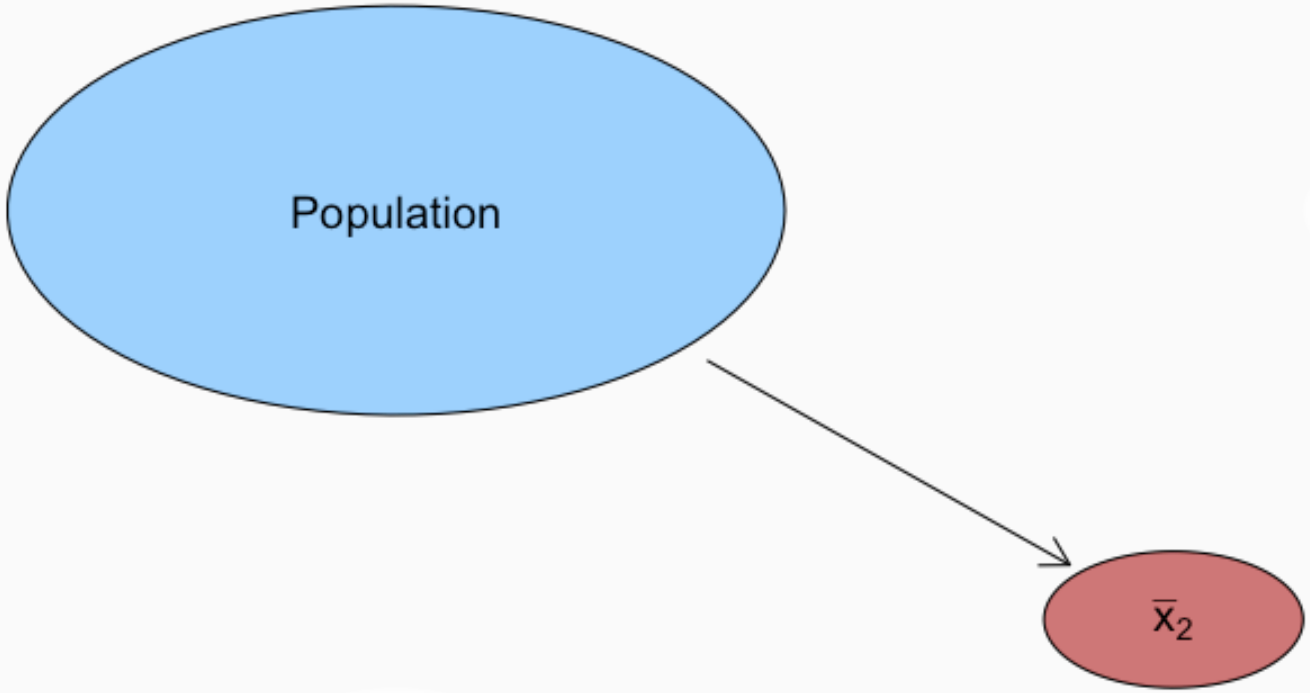

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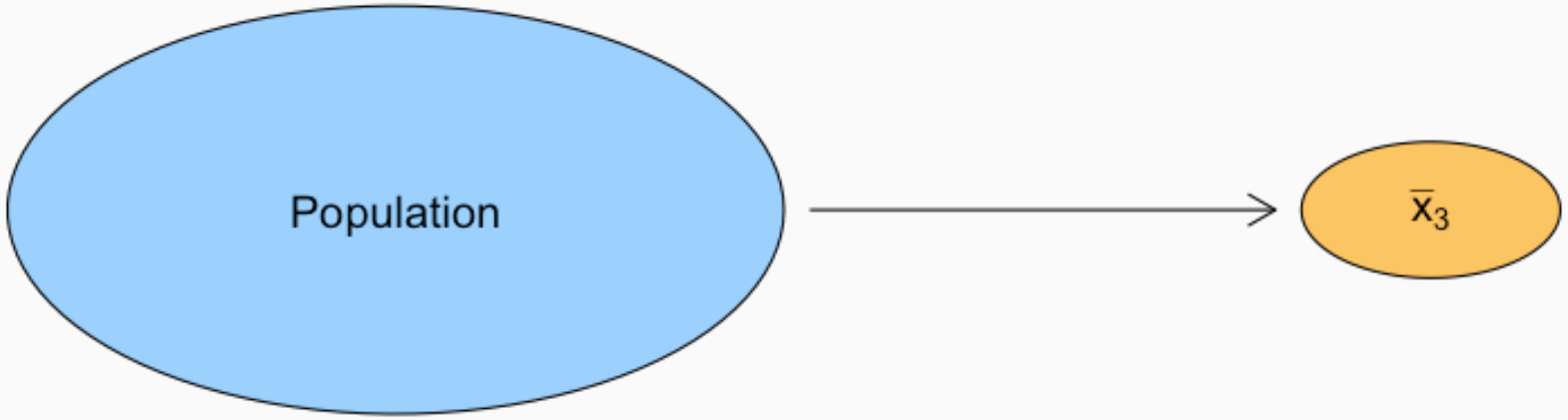

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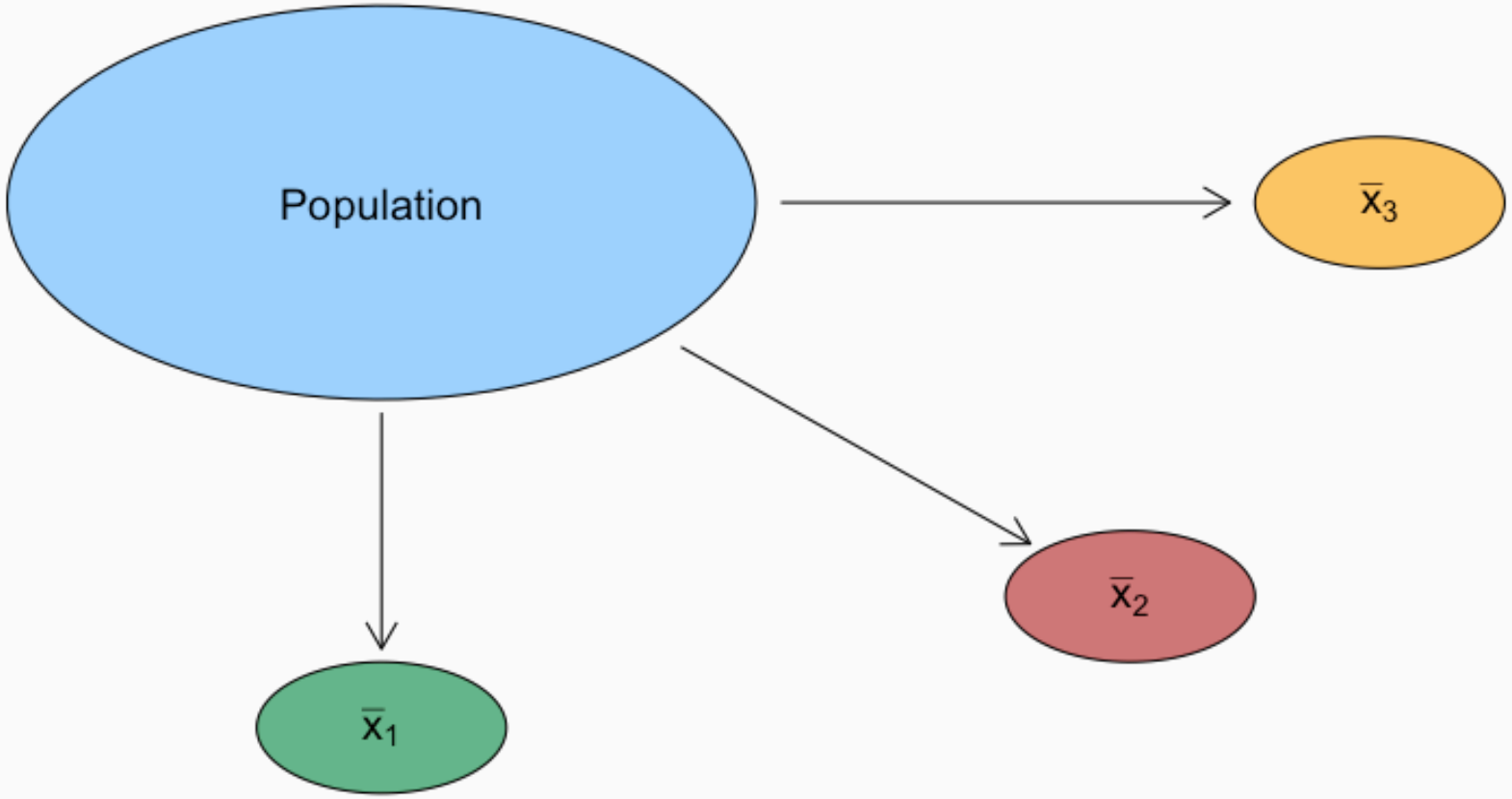
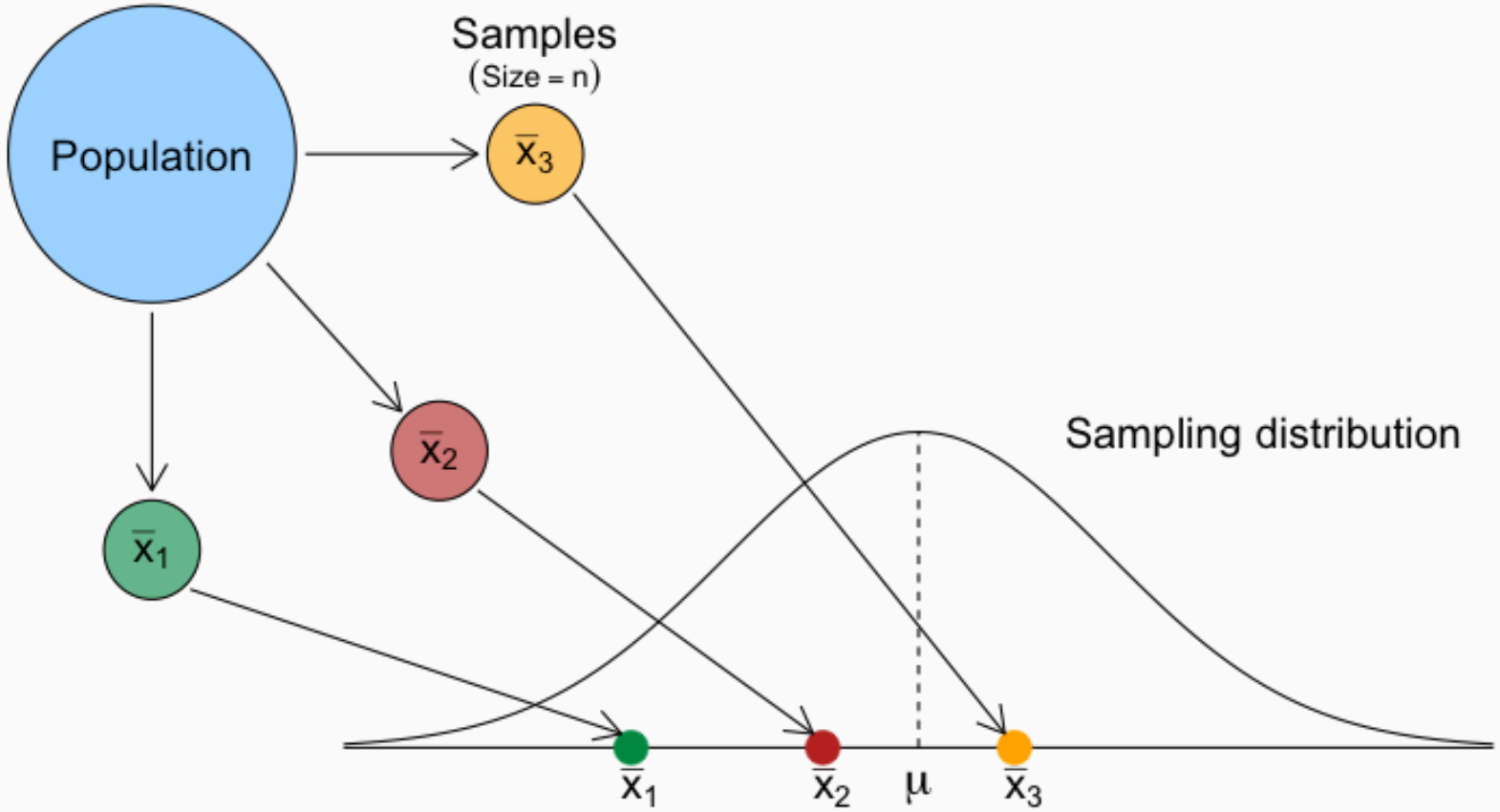


The sampling distribution



The Central Limit Theorem

- There is variability in the estimate of a parameter that depends on the sample (the sampling distribution).
- We would like to have a range of plausible ("common") values for the population parameter
- The shape of the sampling distribution does not depend on the shape of the empirical distribution and, for large samples, can be approximated by a Normal distribution

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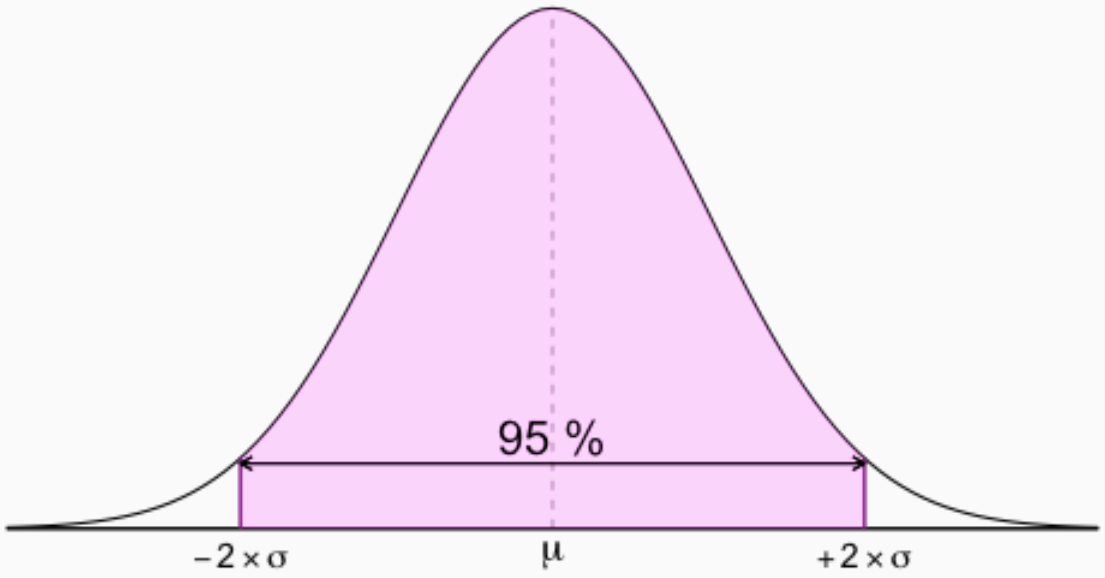
The sampling distribution of the mean is

$$\mathcal{N} = (\mu, \frac{\sigma^2}{n}) \text{ with } \sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}} = \text{standard error (SE)}$$

Let’s put the jigsaw together

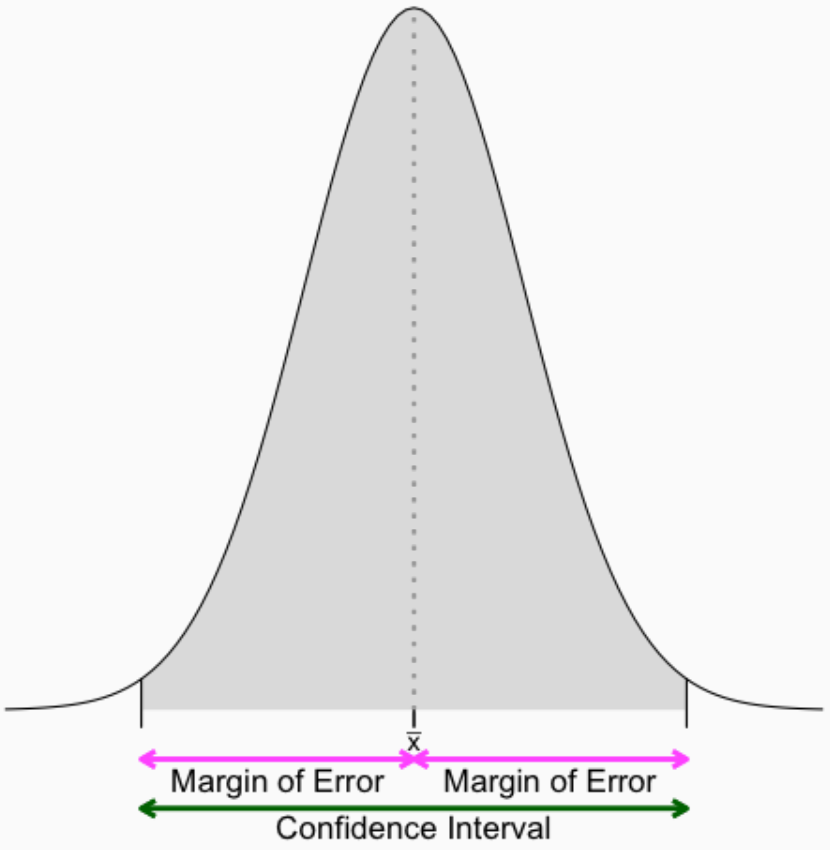
- The sampling distribution can be approximated by a Normal distribution
- In a Normal distribution, about 95% of the values (those "common") lie within two standard deviations of the mean
- A 95% confidence interval (CI) lies (approximately) $\pm 2 \times SE$ from the mean of the sampling distribution



Computing a Confidence Interval

🎯 A 95% CI lies ± 2 × SE from the sampling distribution mean ($\bar{x}$)

1. Compute the SE
2. Compute $2 \times \mathrm{SE}$, *i.e.*,
   95% Margin of Error (ME)
3. Compute the 95% CI as
   $(\bar{x} - \mathrm{ME}\,;\ \bar{x} + \mathrm{ME})$



Confidence intervals for the mean

📌 $\mathcal{N} = (\mu, \frac{\sigma^2}{n})$ with

$$\sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}} = \text{standard error (SE)}$$

$$\sigma = ? \longrightarrow \hat{SE} = \frac{s}{\sqrt{n}}$$

Exercise #6

In a sample of 403 Italian 11-year-old girls, the mean BMI is $18.4 \pm 3.3\ \mathrm{kg/m^2}$

? What is the 95% CI for the true population mean?

05:00

Exercise #6 -- Solution

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1. $\hat{\mathrm{SE}} = s/\sqrt{n} = 3.3/\sqrt{403} = 0.16$

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1. $\hat{\mathrm{SE}} = s/\sqrt{n} = 3.3/\sqrt{403} = 0.16$

2. $95\%\ \mathrm{ME} = 2 \times \hat{\mathrm{SE}} = 2 \times 0.16 = 0.32$

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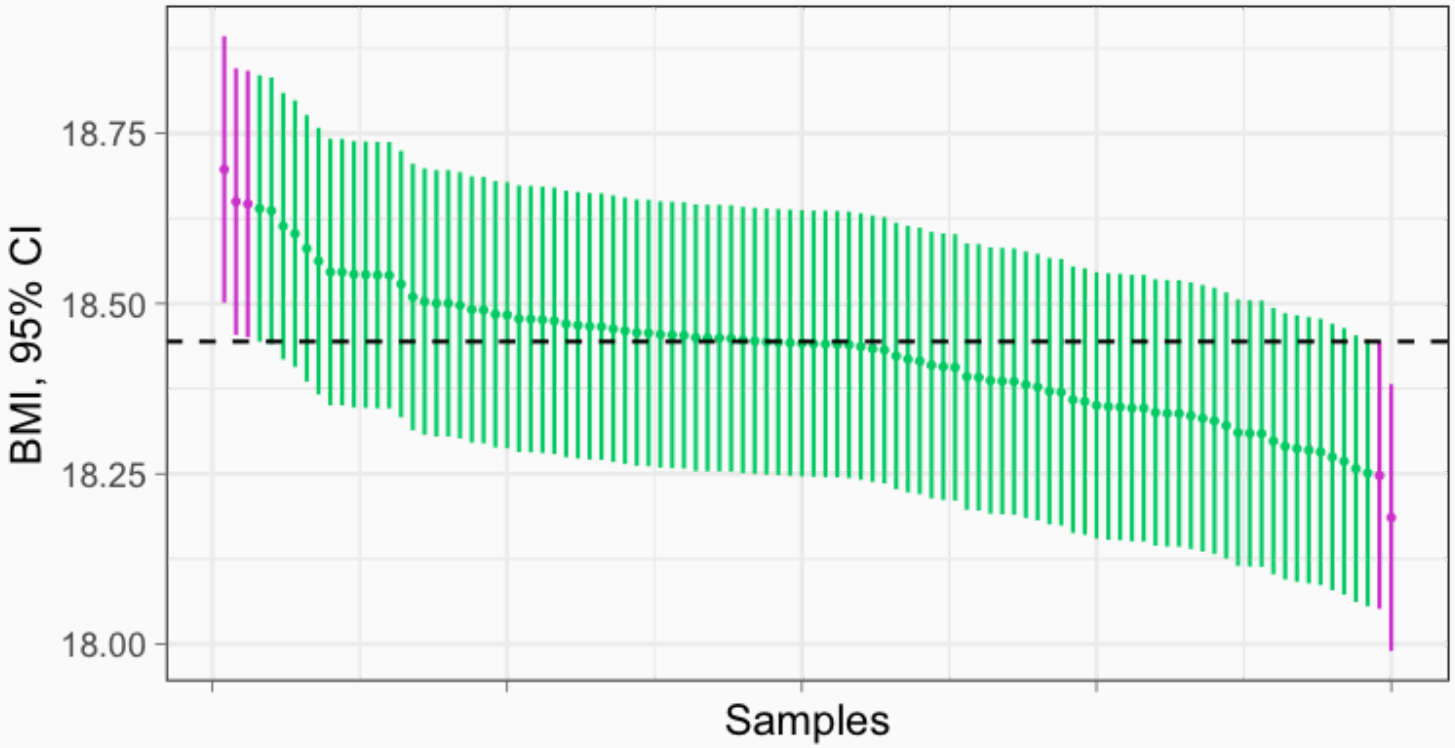
2. $95\%\ \mathrm{ME} = 2 \times \hat{\mathrm{SE}} = 2 \times 0.16 = 0.32$

3. $$\begin{aligned} 95\%\mathrm{CI} &= (\bar{x} - 95\%\ \mathrm{ME}\ ;\ \bar{x} + 95\%\ \mathrm{ME}) = \\ &= (18.4 - 0.32\ ;\ 18.4 + 0.32) = \\ &= (18.08\ ;\ 18.72) \end{aligned}$$

Interpreting confidence intervals

- If we could sample the population 100 times and compute the corresponding 100 CIs, 95 of them would include the true population parameter

📌 Population: Italian 11-year-old girls $\mu = 18.4\ \mathrm{kg/m^2}$



Exercise #7

? In the Results section, the authors reported the following

In Mexican Americans, the mean age at menarche was 12.09 years (95% CI = 11.81 to 12.37 years old)

This means that...

a) The age at menarche for Mexican American girls lies between 11.81 and 12.37 years old
b) 95% of Mexican American girls experience menarche between 11.81 and 12.37 years old
c) The mean age at menarche for Mexican American girls has a 95% probability of being between 11.81 and 12.37 years.

00:30

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The confidence on the confidence intervals

- A confidence interval is a range of values which includes the estimated parameter with a given level of confidence (*e.g.*, 95%)
- Its width depends on the margin of error
- The margin of error is evaluated as:

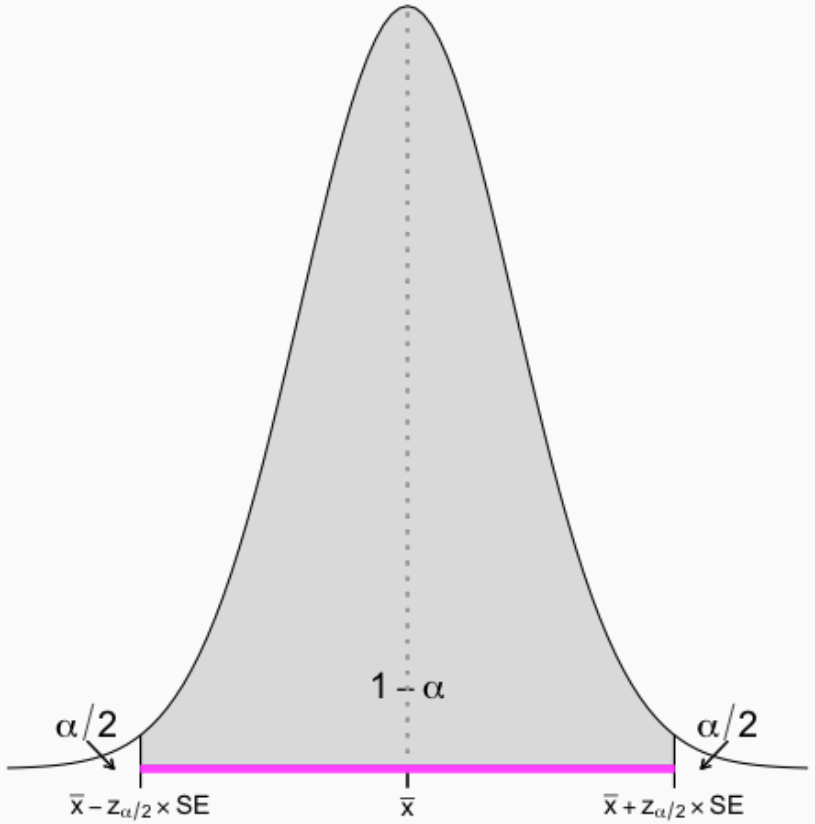
$$2 \times \mathrm{SE} \text{ (at a confidence level of 95\%)}$$

But why do we multiply by 2?

The confidence level

🎯 We multiply by the critical value from the Normal distribution z that corresponds to our confidence level

Confidence Level	α	$\alpha/2$
95%	5%	2.5%

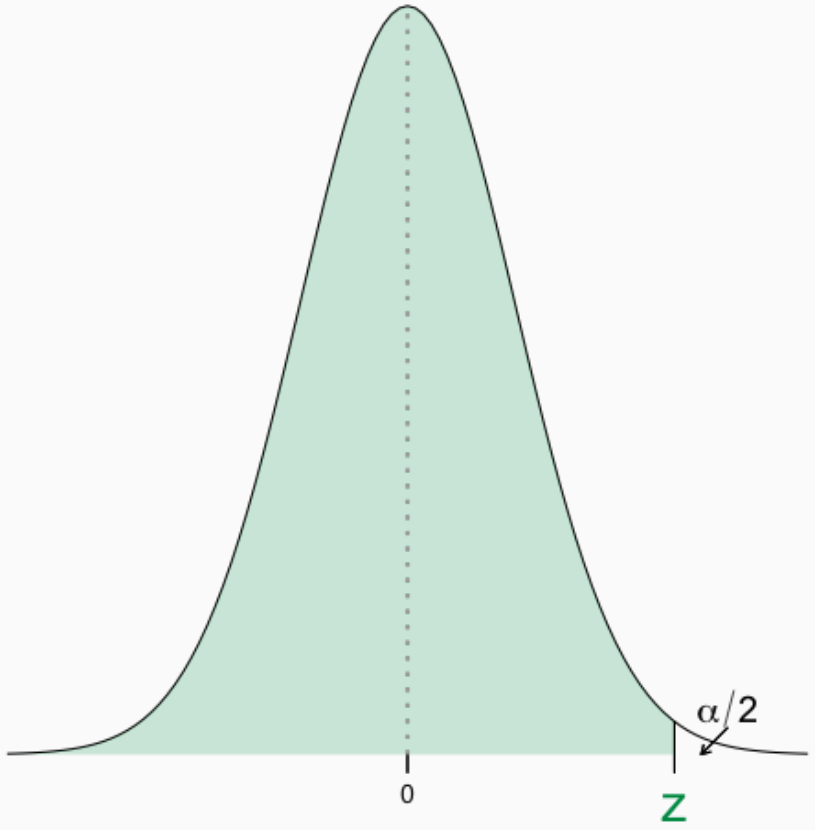


The confidence level

🎯 We multiply by the critical value from the Normal distribution z that corresponds to our confidence level

Confidence Level	α	$\alpha/2$
95%	5%	2.5%

$$100\% - 2.5\% = 97.5\%$$



The confidence level

🎯 We multiply by the critical value from the Normal distribution z that corresponds to our confidence level

Confidence Level	α	$\alpha/2$
95%	5%	2.5%

$$100\% - 2.5\% = 97.5\% \rightarrow z = 1.96$$

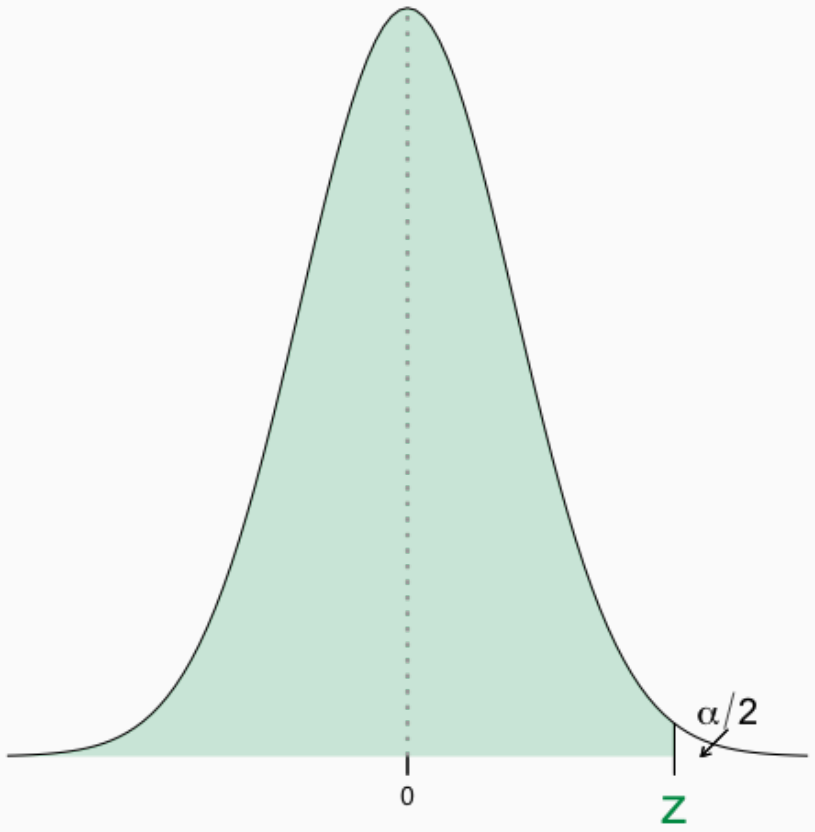
z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817

The confidence level

🎯 We multiply by the critical value from the Normal distribution z that corresponds to our confidence level

Confidence Level	α	$\alpha/2$
95%	5%	2.5%
90%	10%	5.0%
99%	1%	0.5%

$$100\% - 2.5\% = 97.5\% \rightarrow z = 1.96$$
$$100\% - 5.0\% = 95.0\% \rightarrow z = 1.65$$
$$100\% - 0.5\% = 99.5\% \rightarrow z = 2.58$$



Exercise #8

? Given that $\mathcal{N} = (\mu, \frac{\sigma^2}{n})$ with $\sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}} \rightarrow$ standard error (SE), how can one reduce the width of the confidence interval?

a) increasing n
b) decreasing n
c) increasing σ
d) decreasing σ
e) increasing the confidence level
f) decreasing the confidence level

00:30

Exercise #8 -- Solution

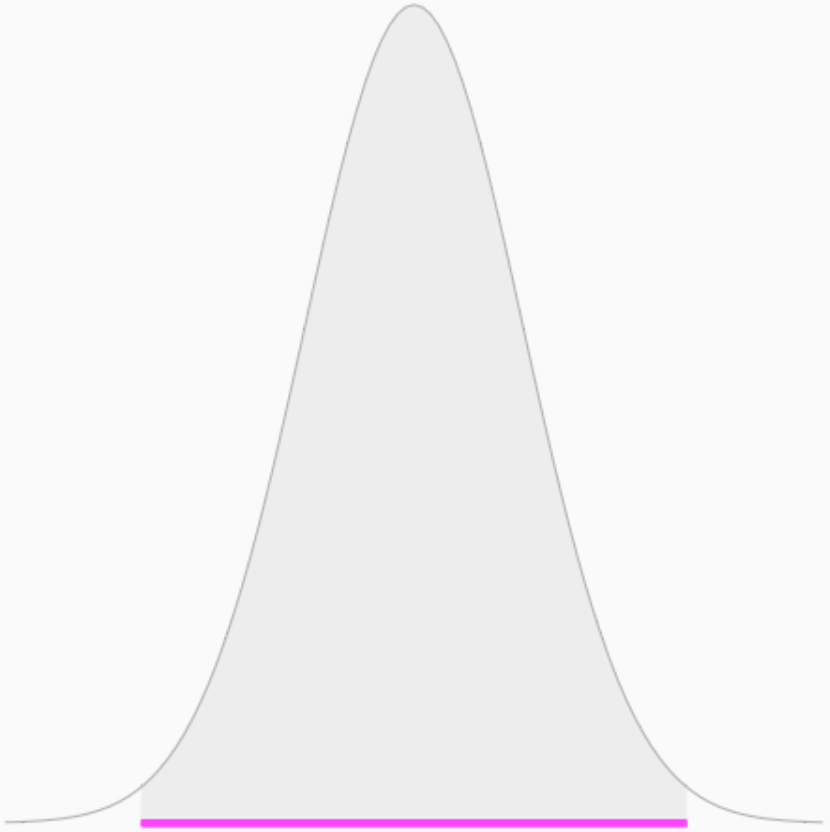
❓ Given that $\mathcal{N} = (\mu, \frac{\sigma^2}{n})$ with $\sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}} \rightarrow$ standard error (SE), how can one reduce the width of the confidence interval?

a) increasing n ✅
b) decreasing n
c) increasing σ
d) decreasing σ ☑️
e) increasing the confidence level
f) decreasing the confidence level ✅

Exercise #9

? If the CI is large we are...

a) more likely of including μ
b) less likely of including μ
c) there is no difference



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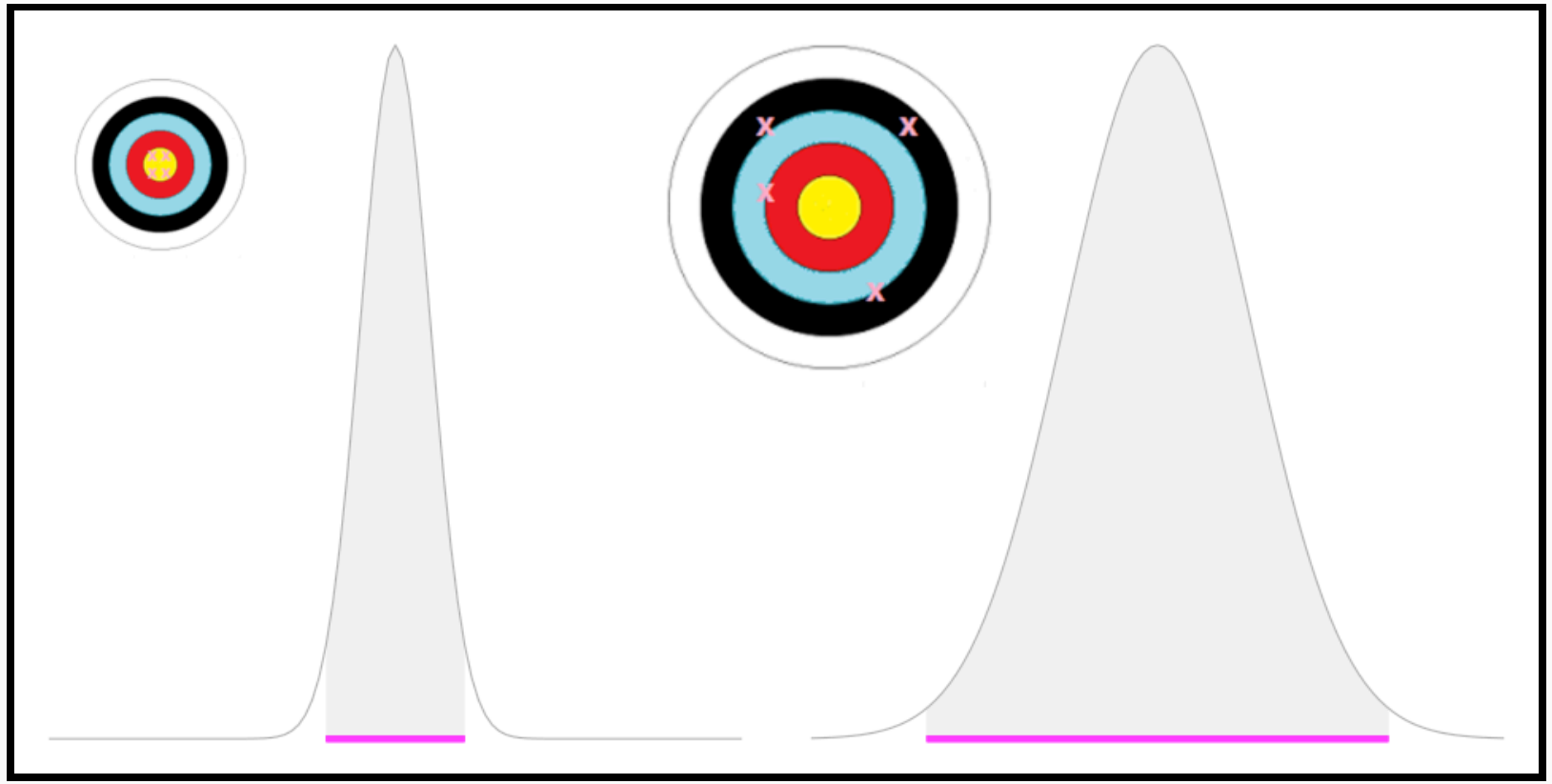
Exercise #9 -- Solution



Exercise #9 -- Solution

❓ If the CI is large we are...

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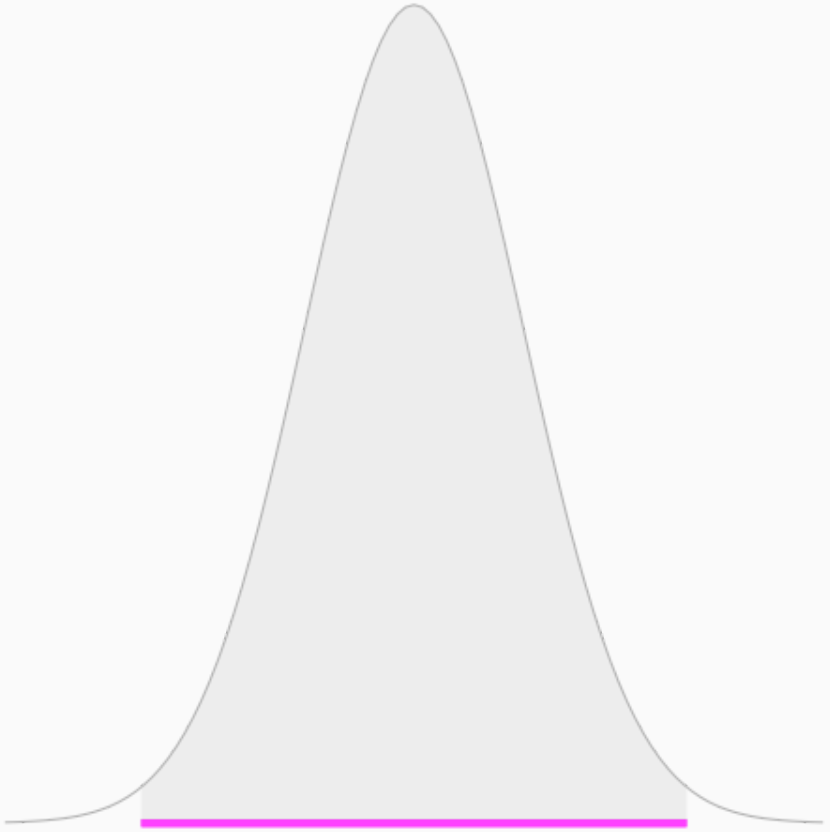
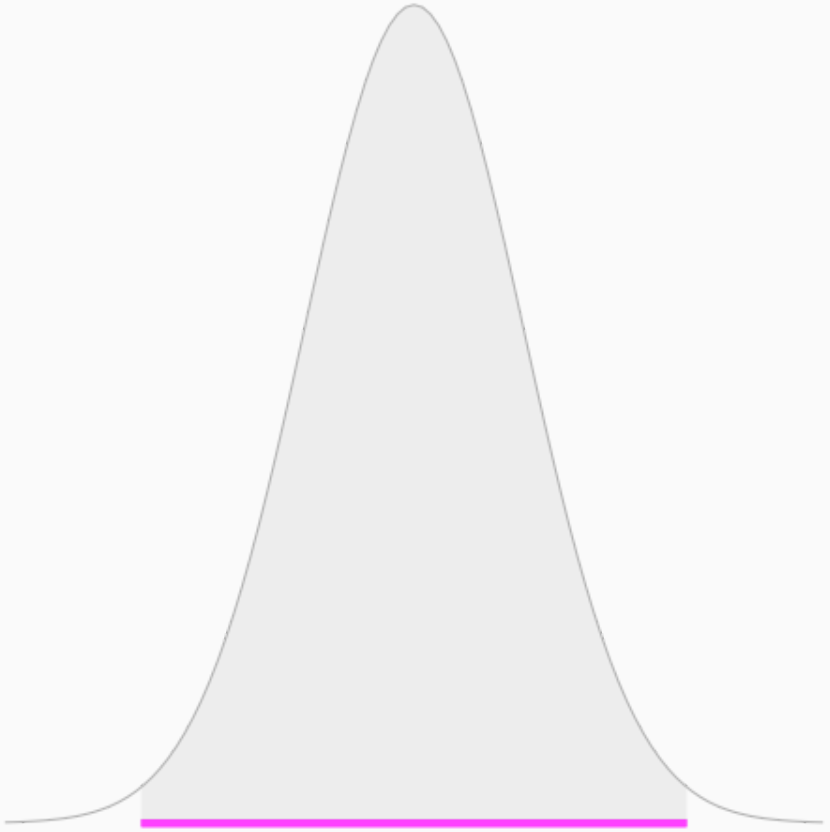


Exercise #10

? If the CI is large we are...

a) more precise
b) less precise
c) there is no difference



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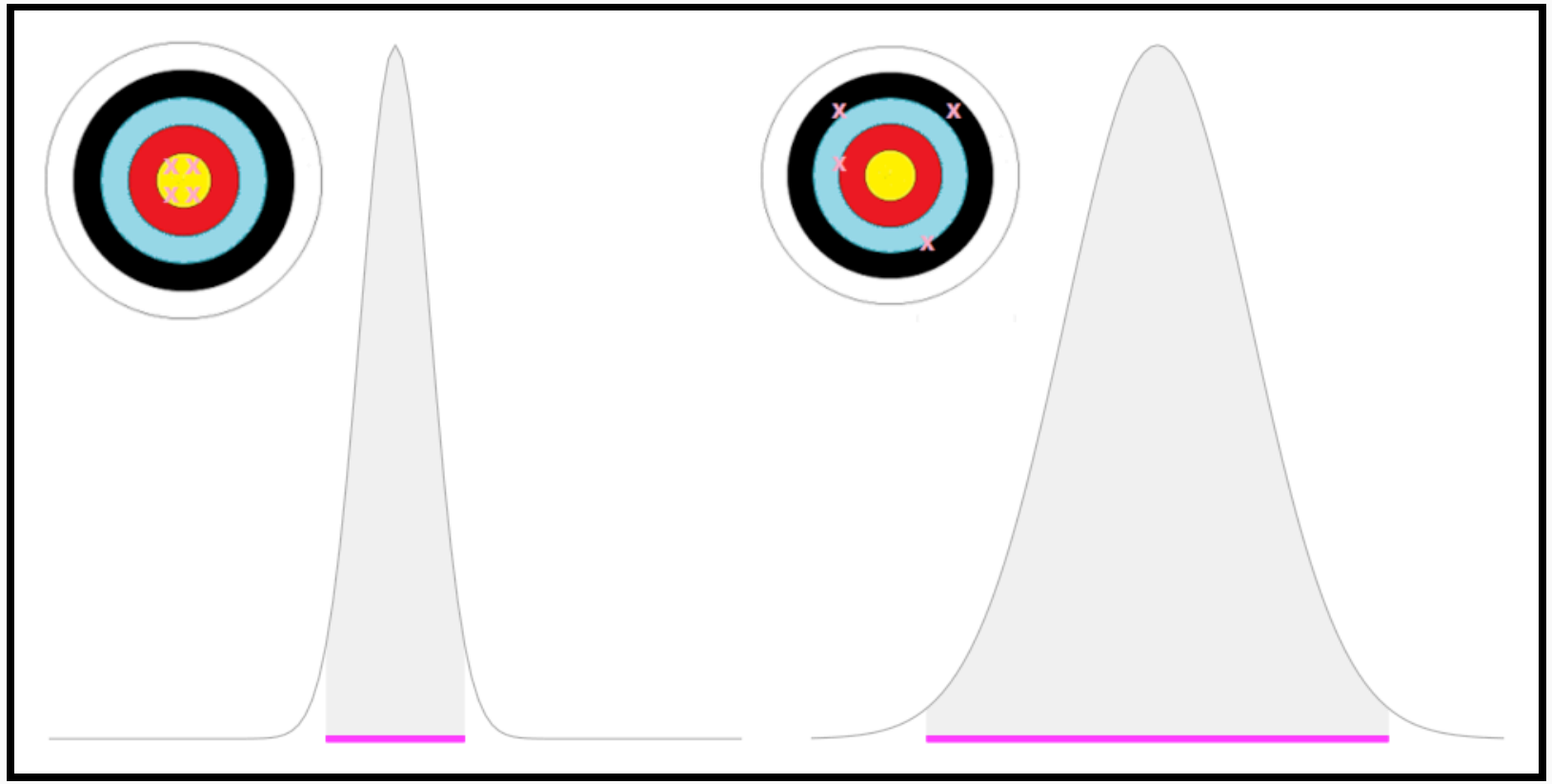
Exercise #10 -- Solution



Exercise #10 -- Solution

❓ If the CI is large we are...

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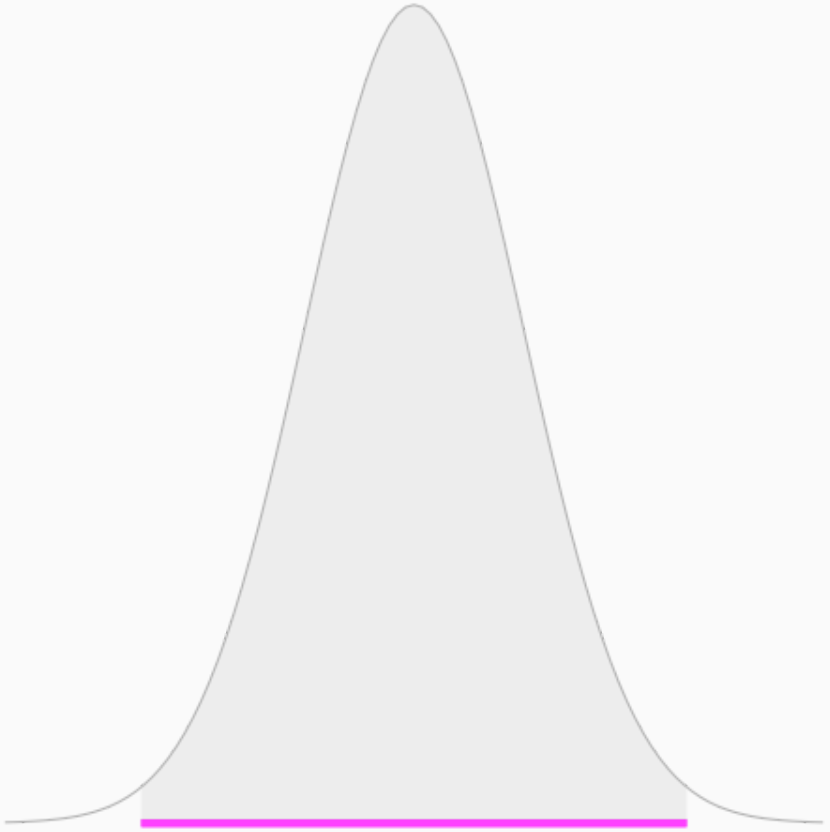
c) there is no difference



The Central Limit Theorem

- There is variability in the estimate of a parameter that depends on the sample (the sampling distribution).
- We would like to have a range of plausible ("common") values for the population parameter
- The shape of the sampling distribution does not depend on the shape of the empirical distribution and, **for large samples**, can be approximated by a Normal distribution

How large is “large”?

The Central Limit Theorem

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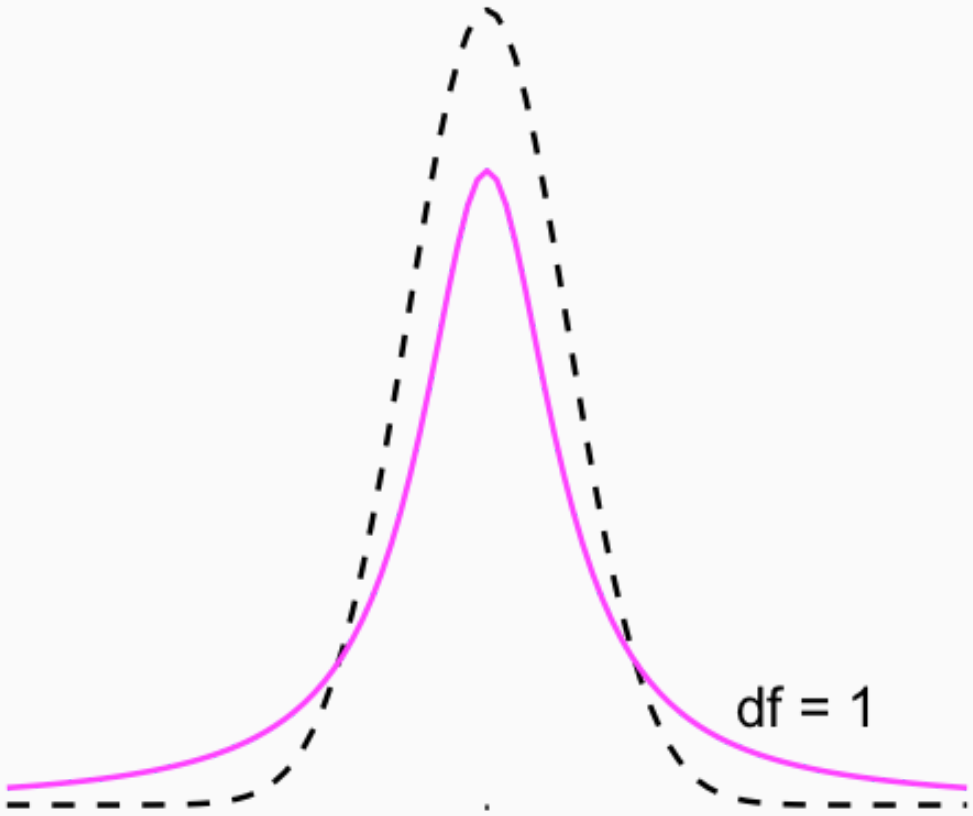
What if our sample is "small"?

What if our sample is "small"?

- Cannot approximated with a Normal distribution
- One should use the Student's t distribution

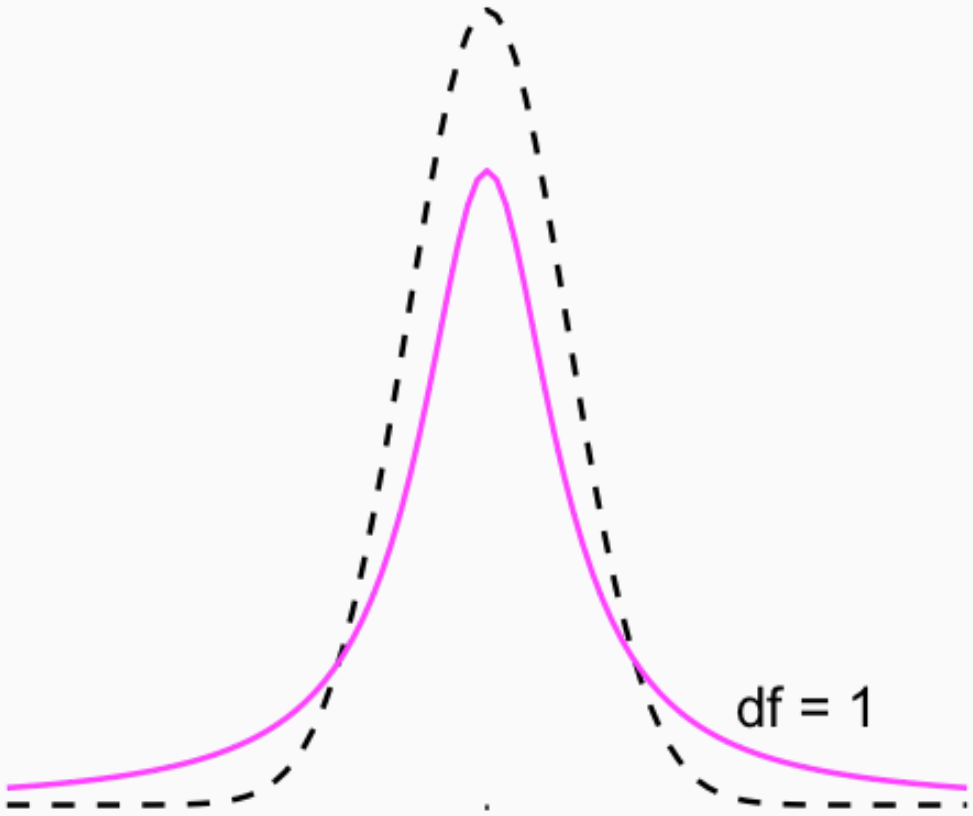
The Student's t distribution

- It takes into account the degrees of freedom (df)



The Student's t distribution

- It takes into account the degrees of freedom (df)
  - In a sample of size $n \rightarrow \mathrm{df} = n - 1$
  - In two samples of size n_1 and $n_2 \rightarrow \mathrm{df} = n_1 + n_2 - 2$



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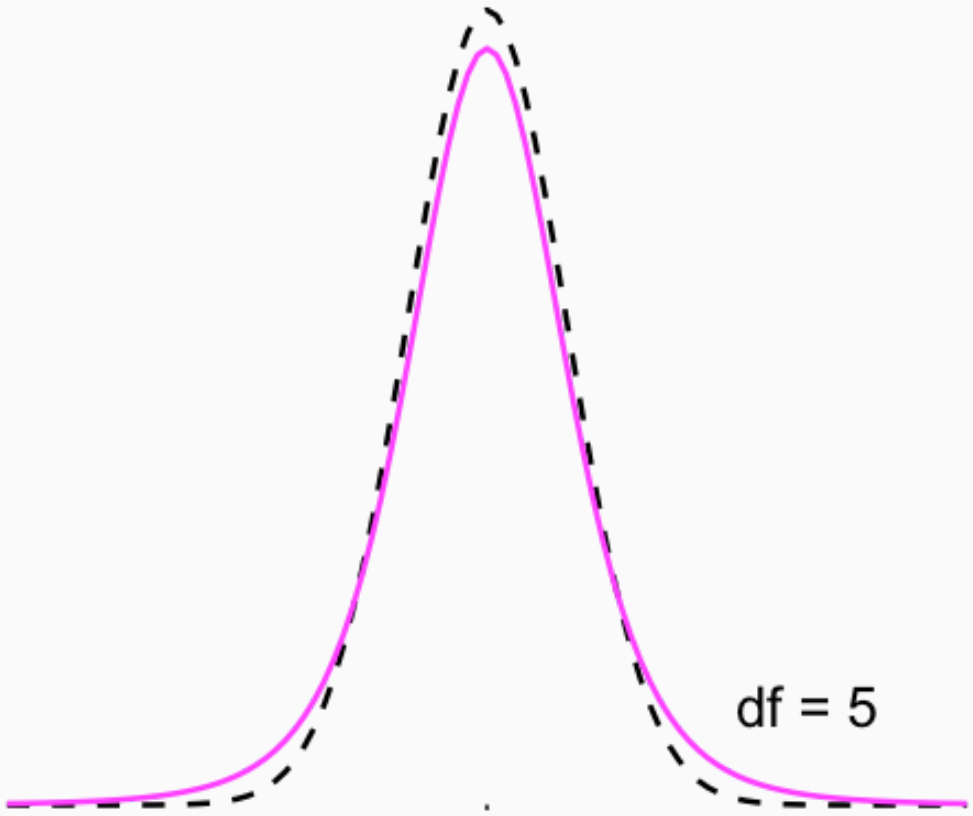

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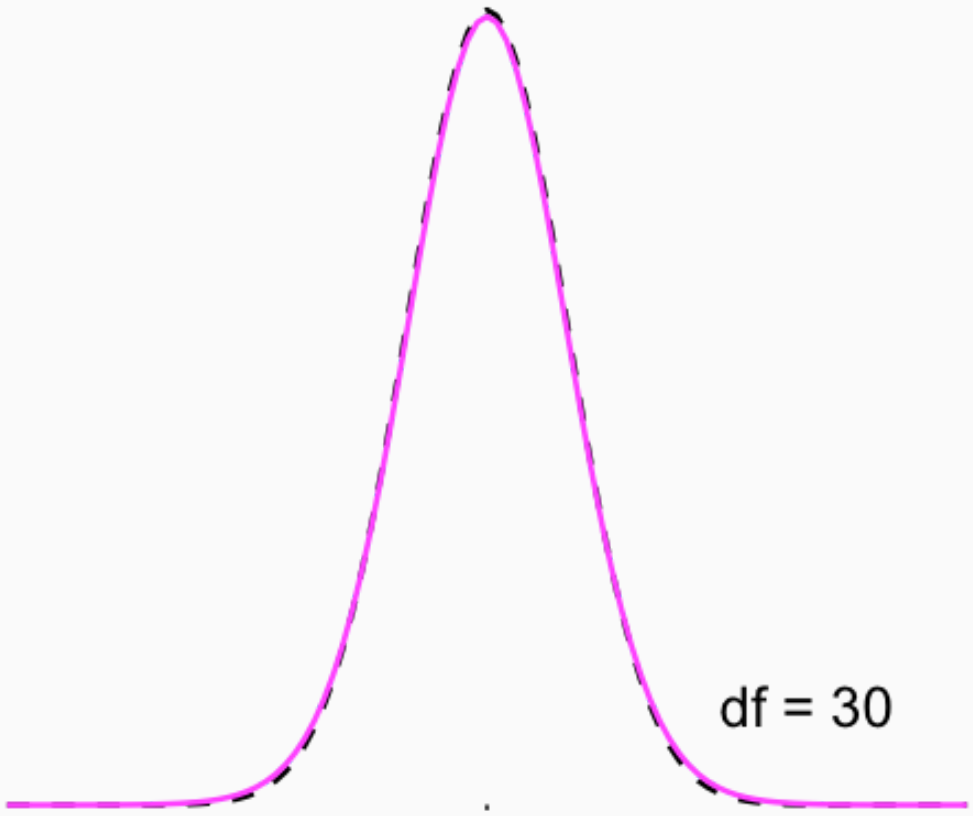

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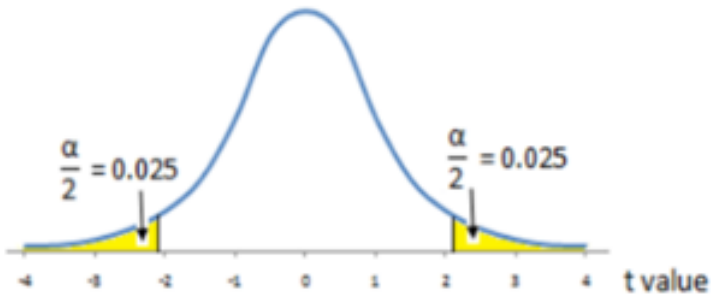
$$\longrightarrow 95\% \text{ ME} = t \times \hat{SE}$$



Confidence	90	95	95.45	99	99.73
alpha	0.1000	0.0500	0.0455	0.0100	0.0027
df					
1	6.314	12.706	13.968	63.657	235.784
2	2.920	4.303	4.527	9.925	19.206
3	2.353	3.182	3.307	5.841	9.219
4	2.132	2.776	2.869	4.604	6.620
29	1.699	2.045	2.090	2.756	3.280
30	1.697	2.042	2.087	2.750	3.270
40	1.684	2.021	2.064	2.704	3.199
50	1.676	2.009	2.051	2.678	3.157
60	1.671	2.000	2.043	2.660	3.130
70	1.667	1.994	2.036	2.648	3.111
80	1.664	1.990	2.032	2.639	3.096
90	1.662	1.987	2.028	2.632	3.085
100	1.660	1.984	2.025	2.626	3.077
1000	1.646	1.962	2.003	2.581	3.007
∞	1.645	1.960	2.000	2.576	3.000

Exercise #11

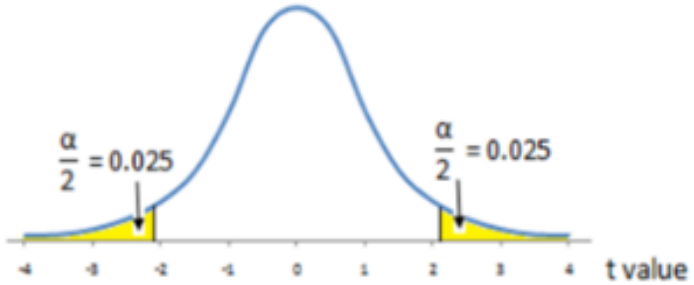
?

$$n = 15$$
$$\bar{x}_{\text{BMI}} = 25.0\ \text{kg/m}^2$$
$$s_{\text{BMI}} = 2.7\ \text{kg/m}^2$$

$$\hat{\text{SE}} = s/\sqrt{n} = \ ?$$
$$\text{df} = n - 1 = \ ?$$

$$95\%\ \text{CI} = \ ?$$

05:00



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6	1.943	2.447	2.517	3.707	4.904
7	1.895	2.365	2.429	3.499	4.530
8	1.860	2.306	2.366	3.355	4.277
9	1.833	2.262	2.320	3.250	4.094
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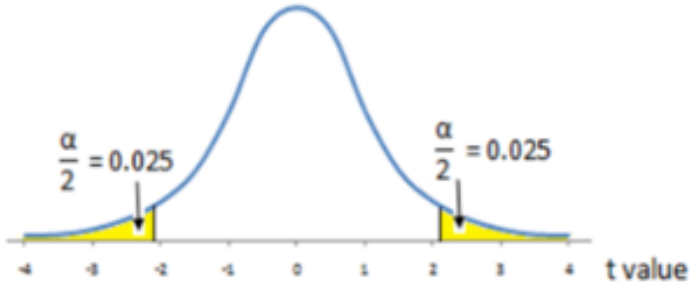
11	1.796	2.201	2.255	3.106	3.850
12	1.782	2.179	2.231	3.055	3.764
13	1.771	2.160	2.212	3.012	3.694
14	1.761	2.145	2.195	2.977	3.636
15	1.753	2.131	2.181	2.947	3.586
16	1.746	2.120	2.169	2.921	3.544
17	1.740	2.110	2.158	2.898	3.507

Exercise #11 -- Solution

?

$$n = 15$$
$$\bar{x}_{\text{BMI}} = 25.0\ \text{kg/m}^2$$
$$s_{\text{BMI}} = 2.7\ \text{kg/m}^2$$

$$\hat{\text{SE}} = s/\sqrt{n} = \frac{2.7}{\sqrt{15}} = 0.7$$
$$\text{df} = n - 1 = 15 - 1 = 14$$



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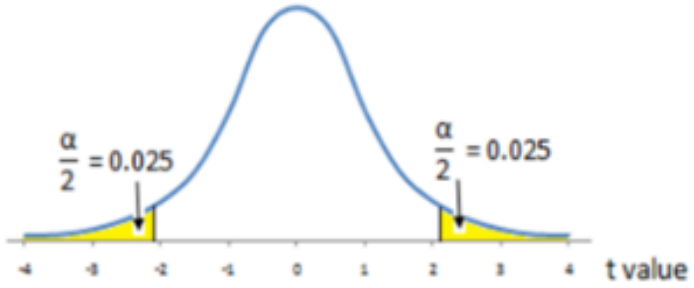
$$\hat{\text{SE}} = s/\sqrt{n} = \frac{2.7}{\sqrt{15}} = 0.7$$
$$\text{df} = n - 1 = 15 - 1 = 14$$

$$95\%\ \text{ME} = t \times \hat{SE} =$$
$$= 2.145 \times 0.7 = 1.5$$



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$$n = 15$$
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$$s_{\text{BMI}} = 2.7\ \text{kg/m}^2$$

$$\hat{\text{SE}} = s/\sqrt{n} = \frac{2.7}{\sqrt{15}} = 0.7$$
$$\text{df} = n - 1 = 15 - 1 = 14$$

$$95\%\ \text{ME} = 2.145 \times 0.7 = 1.5$$

$$\begin{aligned} 95\%\ \text{CI} &= (\bar{x} - 95\%\ \text{ME}\ ;\ \bar{x} + 95\%\ \text{ME}) = \\ &= (25.0 - 1.5;\ 25.0 + 1.5) = \\ &= (23.5; 26.5) \end{aligned}$$

Confidence intervals for differences of means

📌 $$\mathcal{N} = (\mu_{\mathrm{i}} - \mu_{\mathrm{c}}, \sqrt{\frac{\sigma_{\mathrm{i}}^2}{n_{\mathrm{i}}} + \frac{\sigma_{\mathrm{c}}^2}{n_{\mathrm{c}}}})$$

$$\hat{\mathrm{SE}} = \sqrt{\frac{s_{\mathrm{i}}^2}{n_{\mathrm{i}}} + \frac{s_{\mathrm{c}}^2}{n_{\mathrm{c}}}}$$

Exercise #12

? Which is the range of values which includes the *true* difference in mean between the two groups with a 95% confidence level?

Interventions Twenty mg of dexamethasone intravenously daily for 5 days, 10 mg of dexamethasone daily for 5 days or until ICU discharge, plus standard care (n=151) or standard care alone (n=148).

Results A total of 299 patients (mean [SD] age, 61 [14] years; 37% women) were enrolled and all completed follow-up. Patients randomized to the dexamethasone group had a mean 6.6 ventilator-free days (95% CI, 5.0-8.2) during the first 28 days vs 4.0 ventilator-free days (95% CI, 2.9-5.4) in the standard care group

$$n_\mathrm{i} = ?, \quad \bar{x}_\mathrm{i} = ?, \quad s_\mathrm{i} = 10.0$$
$$n_\mathrm{c} = ?, \quad \bar{x}_\mathrm{c} = ?, \quad s_\mathrm{c} = 8.7$$

$$\hat{\mathrm{SE}} = \sqrt{\frac{s_\mathrm{i}^2}{n_\mathrm{i}} + \frac{s_\mathrm{c}^2}{n_\mathrm{c}}} = ?$$

05:00

Tomazini B.M., *et al.*, *Effect of Dexamethasone on Days Alive and Ventilator-Free in Patients With Moderate or Severe Acute Respiratory Distress Syndrome and COVID-19*, JAMA, 2020

Exercise #12 -- Solution

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$$n_{\mathrm{i}} = 151, \quad \bar{x}_{\mathrm{i}} = 6.6, \quad s_{\mathrm{i}} = 10.0$$
$$n_{\mathrm{c}} = 148, \quad \bar{x}_{\mathrm{c}} = 4.0, \quad s_{\mathrm{c}} = 8.7$$

$$\hat{\mathrm{SE}} = \sqrt{\frac{s_{\mathrm{i}}^2}{n_{\mathrm{i}}} + \frac{s_{\mathrm{c}}^2}{n_{\mathrm{c}}}} = \sqrt{\frac{10.0^2}{151} + \frac{8.7^2}{148}} = 1.08$$

Exercise #12 -- Solution

? Which is the range of values which includes the *true* difference in mean between the two groups with a 95% confidence level?

$$n_{\mathrm{i}} = 151, \quad \bar{x}_{\mathrm{i}} = 6.6, \quad s_{\mathrm{i}} = 10.0$$
$$n_{\mathrm{c}} = 148, \quad \bar{x}_{\mathrm{c}} = 4.0, \quad s_{\mathrm{c}} = 8.7$$

$$\hat{\mathrm{SE}} = \sqrt{\frac{s_{\mathrm{i}}^2}{n_{\mathrm{i}}} + \frac{s_{\mathrm{c}}^2}{n_{\mathrm{c}}}} = \sqrt{\frac{10.0^2}{151} + \frac{8.7^2}{148}} = 1.08$$

$$95\%\ \mathrm{ME} = 2 \times \hat{SE} = 2 \times 1.08 = 2.16$$

Exercise #12 -- Solution

? Which is the range of values which includes the *true* difference in mean between the two groups with a 95% confidence level?

$$n_\mathrm{i} = 151, \quad \bar{x}_\mathrm{i} = 6.6, \quad s_\mathrm{i} = 10.0$$
$$n_\mathrm{c} = 148, \quad \bar{x}_\mathrm{c} = 4.0, \quad s_\mathrm{c} = 8.7$$

$$\hat{\mathrm{SE}} = \sqrt{\frac{s_\mathrm{i}^2}{n_\mathrm{i}} + \frac{s_\mathrm{c}^2}{n_\mathrm{c}}} = \sqrt{\frac{10.0^2}{151} + \frac{8.7^2}{148}} = 1.08$$

$$95\%\ \mathrm{ME} = 2 \times \hat{SE} = 2 \times 1.08 = 2.16$$

$$\begin{aligned} 95\%\ \mathrm{CI} &= (\bar{x}_\mathrm{i} - \bar{x}_\mathrm{c}) - 95\%\ \mathrm{ME}\ ;\ (\bar{x}_\mathrm{i} - \bar{x}_\mathrm{c}) + 95\%\ \mathrm{ME} = \\ &= ((6.6 - 4.0) - 2.16\ ;\ (6.6 - 4.0) + 2.16) = \\ &= (2.6 - 2.16\ ;\ 2.6 + 2.16) = (0.44\ ;\ 4.78) \end{aligned}$$

Confidence intervals for proportions

📌 $\mathcal{N} = (\pi, \frac{\pi \times (1-\pi)}{n})$

$\hat{SE} = \sqrt{\frac{\bar{p} \times (1-\bar{p})}{n}}$, where $\bar{p} = \frac{m}{n}$

Exercise #13

? Which is the range of values which includes *true* proportion of women with endometriosis in the population with a 95% confidence level?

A total of 1291 women without a previous diagnosis of endometriosis were included in the study. On the basis of the symptoms, 108 women were referred to a gynecologist. After gynecological examination and transvaginal ultrasound, endometriosis was suspected in 51 women (47.2%). The diagnosis of endometriosis was confirmed by radiological investigations and/or surgery in 46 patients;

$$n = \ ?, \quad m = \ ?$$
$$\bar{p} = \frac{m}{n} = \ ?$$

$$\hat{SE} = \sqrt{\frac{\bar{p} \times (1 - \bar{p})}{n}} = \ ?$$

05:00

Ferrero S.., *et al.*, *Prevalence of newly diagnosed endometriosis in women attending the general practitioner*, Int J Gynaecol Obstet, 2020

Exercise #13 -- Solution

? Which is the range of values which includes *true* proportion of women with endometriosis in the population with a 95% confidence level?

> A total of 1291 women without a previous diagnosis of endometriosis were included in the study. On the basis of the symptoms, 108 women were referred to a gynecologist. After gynecological examination and transvaginal ultrasound, endometriosis was suspected in 51 women (47.2%). The diagnosis of endometriosis was confirmed by radiological investigations and/or surgery in 46 patients;

$$n = 1291, \quad m = 46$$

$$\bar{p} = \frac{m}{n} = \frac{46}{1291} = 0.036 = 3.6\%$$

$$\hat{SE} = \sqrt{\frac{\bar{p} \times (1-\bar{p})}{n}} = \sqrt{\frac{0.036 \times (1-0.036)}{1291}} = 0.005$$

Exercise #13 -- Solution

? Which is the range of values which includes *true* proportion of women with endometriosis in the population with a 95% confidence level?

$$n = 1291, \quad m = 46$$
$$\bar{p} = \frac{m}{n} = \frac{46}{1291} = 0.036 = 3.6\%$$

$$\hat{SE} = \sqrt{\frac{\bar{p} \times (1 - \bar{p})}{n}} = \sqrt{\frac{0.036 \times (1 - 0.036)}{1291}} = 0.005$$

$$95\% \text{ ME} = 2 \times \hat{SE} = 2 \times 0.005 = 0.01$$

Exercise #13 -- Solution

? Which is the range of values which includes *true* proportion of women with endometriosis in the population with a 95% confidence level?

$$n = 1291, \quad m = 46$$
$$\bar{p} = \frac{m}{n} = \frac{46}{1291} = 0.036 = 3.6\%$$

$$\hat{SE} = \sqrt{\frac{\bar{p} \times (1-\bar{p})}{n}} = \sqrt{\frac{0.036 \times (1-0.036)}{1291}} = 0.005$$

$$95\% \text{ ME} = 2 \times \hat{SE} = 2 \times 0.005 = 0.01$$

$$\begin{aligned} 95\% \text{ CI} &= (\bar{p} - 95\% \text{ ME} \; ; \; \bar{p} + 95\% \text{ ME} = \\ &= (0.036 - 0.01 \; ; \; 0.036 + 0.01) = \\ &= (0.026 \; ; \; 0.046) = (2.6\% \; ; \; 4.6\%) \end{aligned}$$

Confidence intervals for differences of proportion

📌 $$\mathcal{N} = (\pi_{\mathrm{i}} - \pi_{\mathrm{c}}, \frac{\pi_i \times (1-\pi_i)}{n_i} + \frac{\pi_c \times (1-\pi_c)}{n_c})$$

$$\hat{\mathrm{SE}} = \sqrt{\frac{\bar{p}_i \times (1-\bar{p}_i)}{n_i} + \frac{\bar{p}_c \times (1-\bar{p}_c)}{n_c}}$$

Exercise #14

? Which is the range of values which includes the *true* difference in proportion between the two groups with a 95% confidence level?

> From April 1991 through December 20, 1993, the cutoff date for the first interim analysis of efficacy, 477 pregnant women were enrolled; during the study period, 409 gave birth to 415 live-born infants. HIV-infection status was known for 363 births (180 in the zidovudine group and 183 in the placebo group). Thirteen infants in the zidovudine group and 40 in the placebo group were HIV-infected.

$$n_{\mathrm{i}} = ?, \quad m_{\mathrm{i}} = ?, \quad p_{\mathrm{i}} = \frac{m_{\mathrm{i}}}{n_{\mathrm{i}}} = ?$$
$$n_{\mathrm{c}} = ?, \quad m_{\mathrm{c}} = ?, \quad p_{\mathrm{c}} = \frac{m_{\mathrm{c}}}{n_{\mathrm{c}}} = ?$$

$$\hat{\mathrm{SE}} = \sqrt{\frac{\bar{p}_{\mathrm{i}} \times (1-\bar{p}_{\mathrm{i}})}{n_{\mathrm{i}}} + \frac{\bar{p}_{\mathrm{c}} \times (1-\bar{p}_{\mathrm{c}})}{n_{\mathrm{c}}}} = ?$$

05:00

Connor, E.M. *et al.*, *Reduction of Maternal-Infant Transmission of Human Immunodeficiency Virus Type 1 with Zidovudine Treatment*, NEJM, 1994

Exercise #14 -- Solution

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$$n_{\mathrm{i}} = 180, \quad m_{\mathrm{i}} = 13, \quad p_{\mathrm{i}} = \frac{m_{\mathrm{i}}}{n_{\mathrm{i}}} = \frac{13}{180} = 0.07$$
$$n_{\mathrm{c}} = 183, \quad m_{\mathrm{c}} = 40, \quad p_{\mathrm{c}} = \frac{m_{\mathrm{c}}}{n_{\mathrm{c}}} = \frac{40}{183} = 0.22$$

$$\hat{\mathrm{SE}} = \sqrt{\frac{\bar{p}_{\mathrm{i}} \times (1-\bar{p}_{\mathrm{i}})}{n_{\mathrm{i}}} + \frac{\bar{p}_{\mathrm{c}} \times (1-\bar{p}_{\mathrm{c}})}{n_{\mathrm{c}}}} = \sqrt{\frac{0.07 \times (1-0.07)}{180} + \frac{0.22 \times (1-0.22)}{183}} = 0.036$$

Exercise #14 -- Solution

? Which is the range of values which includes the *true* difference in proportion between the two groups with a 95% confidence level?

$$n_{\mathrm{i}} = 180, \quad m_{\mathrm{i}} = 13, \quad p_{\mathrm{i}} = \frac{m_{\mathrm{i}}}{n_{\mathrm{i}}} = \frac{13}{180} = 0.07$$
$$n_{\mathrm{c}} = 183, \quad m_{\mathrm{c}} = 40, \quad p_{\mathrm{c}} = \frac{m_{\mathrm{c}}}{n_{\mathrm{c}}} = \frac{40}{183} = 0.22$$

$$\hat{\mathrm{SE}} = \sqrt{\frac{\bar{p}_{\mathrm{i}} \times (1-\bar{p}_{\mathrm{i}})}{n_{\mathrm{i}}} + \frac{\bar{p}_{\mathrm{c}} \times (1-\bar{p}_{\mathrm{c}})}{n_{\mathrm{c}}}} = \sqrt{\frac{0.07 \times (1-0.07)}{180} + \frac{0.22 \times (1-0.22)}{183}} = 0.036$$

$$95\%\ \mathrm{ME} = 2 \times \hat{SE} = 2 \times 0.036 = 0.072$$

Exercise #14 -- Solution

? Which is the range of values which includes the *true* difference in proportion between the two groups with a 95% confidence level?

$$n_{\mathrm{i}} = 180, \quad m_{\mathrm{i}} = 13, \quad p_{\mathrm{i}} = \frac{m_{\mathrm{i}}}{n_{\mathrm{i}}} = \frac{13}{180} = 0.07$$
$$n_{\mathrm{c}} = 183, \quad m_{\mathrm{c}} = 40, \quad p_{\mathrm{c}} = \frac{m_{\mathrm{c}}}{n_{\mathrm{c}}} = \frac{40}{183} = 0.22$$

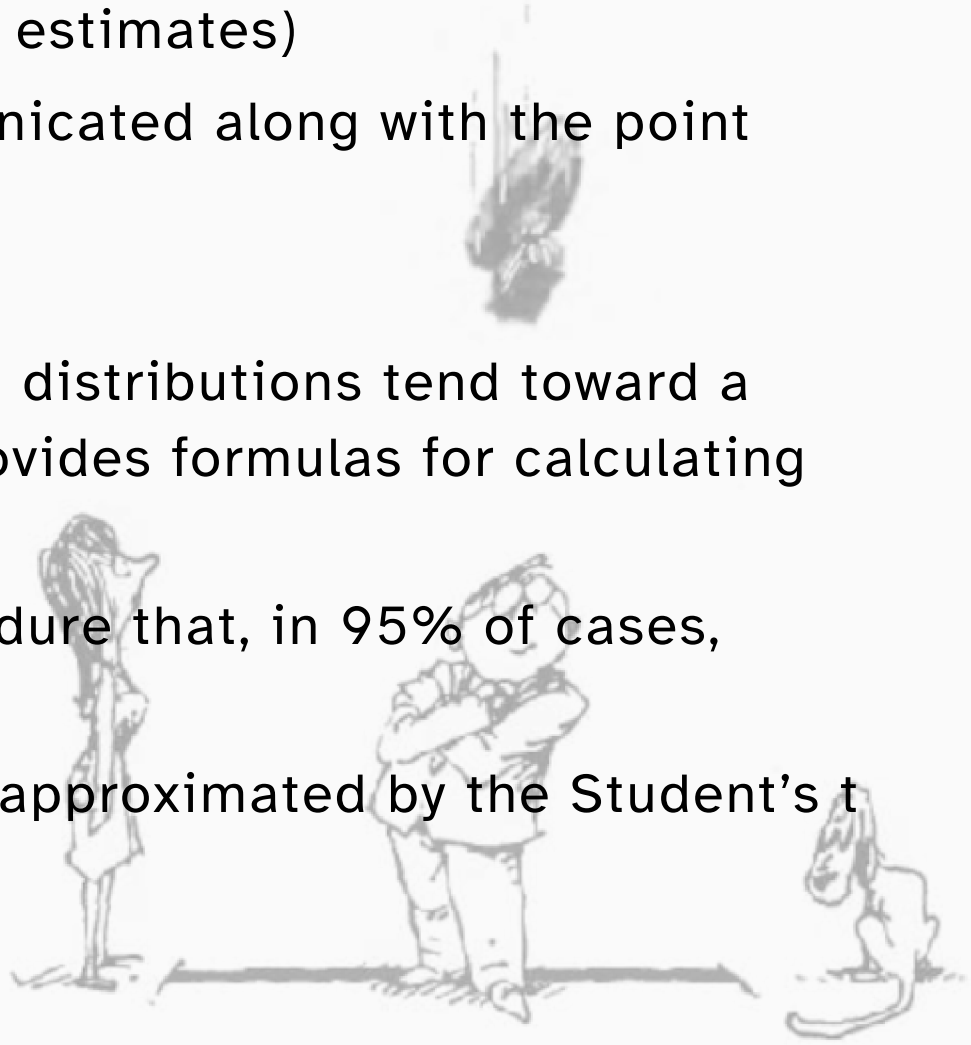
$$\hat{\mathrm{SE}} = \sqrt{\frac{\bar{p}_{\mathrm{i}} \times (1-\bar{p}_{\mathrm{i}})}{n_{\mathrm{i}}} + \frac{\bar{p}_{\mathrm{c}} \times (1-\bar{p}_{\mathrm{c}})}{n_{\mathrm{c}}}} = \sqrt{\frac{0.07 \times (1-0.07)}{180} + \frac{0.22 \times (1-0.22)}{183}} = 0.036$$

$$95\%\ \mathrm{ME} = 2 \times \hat{SE} = 2 \times 0.036 = 0.072$$

$$\begin{aligned} 95\%\ \mathrm{CI} &= ((\bar{p}_{\mathrm{i}} - \bar{p}_{\mathrm{c}}) - 95\%\ \mathrm{ME}\ ;\ (\bar{p}_{\mathrm{i}} - \bar{p}_{\mathrm{c}}) + 95\%\ \mathrm{ME}) = \\ &= (-0.15 - 0.072\ ;\ \ -0.15 + 0.072) = \\ &= (-0.222; -0.078) \end{aligned}$$

Summary

- We use statistics to estimate parameters (point estimates)
- Interval estimates (CI or ME) should be communicated along with the point estimates
- Sample size influences the width of CIs
- The Central Limit Theorem states that sampling distributions tend toward a normal distribution for large samples, and it provides formulas for calculating confidence intervals
- A 95% confidence interval results from a procedure that, in 95% of cases, contains the true population value.
- For small samples, the sampling distribution is approximated by the Student's t distribution



See you tomorrow